Priced Runtime Control over Frozen Agent Harnesses: a Learned Orchestrator, an Off-Policy Promotion Gate, and Drift Metrics

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Abstract

Agent harnesses contain per-request decisions that nobody prices: which model tier answers, whether a sub-result is verified, whether an answer is refined, which specialist is delegated to. The field evolves harnesses at design time; this report studies the complementary runtime problem. We keep the harness frozen and learn a small, cost-priced controller at its decision points, with three pieces: a softmax-linear (or small-LLM) policy trained offline from logged episodes under a budget whose price λ is a learned Lagrange multiplier; an off-policy promotion gate that admits a candidate only when doubly-robust (DR) estimates with bootstrap confidence intervals certify quality and cost; and priced backward/forward-transfer metrics for the harness changing under the controller. Six experiments are reported, with negative results stated as results. On a simulator, over 5 seeds, the gate promotes every first candidate (DR interval covering the ground truth in 4 of 5), drift costs a frozen controller -0.047 ± 0.003 in priced value, and the retrained policy would recover $+0.039 \pm 0.003$ of it, but the gate promotes the retrain in only 2 of 5 seeds, all three reverts being quality false alarms, so the served recovery is $+0.015 \pm 0.009$. On CRMArena-Pro (300 B2B tasks and 150 B2C tasks, every task on three tiers) the per-task oracle beats every fixed tier at lower cost, and the learned per-category policy beats every fixed tier held-out on quality and cost at once in both organisations (B2B: 0.644 at \$0.079 against the frontier tier's 0.614 at \$0.131), while the budgeted controller reaches frontier quality at 22% less cost; at 50 tasks the same policies memorised and lost, and the gate said so. On the realised one-arm log the doubly-robust gate's decision depends on the draw: 30% of redraws promote at 300 tasks (10% at 150, 8% on B2C), and certifying a \$0.005 budget margin remains a thousand-task matter. Two further suites, τ-bench and three stages of EvoHarnessBench, show the same tier-level picture: the mid tier matches or beats the frontier tier at about half its cost and the per-task oracle beats every tier, but a first-step controller has nothing to condition on in τ-bench and does no better than the best fixed arm. A GRPO-trained 1.7B-parameter controller never found the budgeted optimum from near-identical rollout groups; an SFT warm start did the work and GRPO's contribution was the price ($\lambda \approx 0.44$). A planner/worker/verifier decomposition scores below a single mid-tier agent (0.480 vs 0.608), its verifier flags 92% of wrong and 96% of correct episodes, and an orchestration reward model on process features has out-of-fold AUROC 0.469 against 0.821 for the task category alone. Priced self-refinement after a cheap answer is dominated by buying a better tier up front (-0.174 ± 0.085 paired at equal cost). The centrepiece is a continual orchestrator over a frozen multi-agent structure on a simulator calibrated to the real logs, under four pre-registered harness stages over 40 days: the online learner has the lowest regret of every learned arm (1.26 vs 2.60 frozen and 3.03–3.46 for static routers), recovers after every stage, and has bounded, mostly positive orchestrator-level BWT; but a fixed mode (`never_verify`, regret 0.56; 0.74 ± 0.08 against the learner's 1.06 ± 0.21 over 5 seeds) beats it over the whole stream. That headline is a cold-start artefact: the learner spends 12 days on the uniform exploration policy before its first promotion, crosses the fixed mode around day 22–25 and wins every stage window after that (held-out regret from day 25: 0.22 vs 0.50), and its post-crossover edge at the last stage survives 5 seeds ($+0.031 \pm 0.007$). A 5-seed verifier ablation shows the fixed mode's win is itself an artefact of the measured verifier and the 8-call cap: with an informative verifier and a cap the retries do not consume, verification helps ($+0.029 \pm 0.001$ quality for $+18\%$ cost) and `always_verify` becomes the right fixed mode. The gate reverts 32 of 40 candidates (23 false alarms on the quality criterion), and the AWR variant is promoted 14 times on a DR estimate biased by $+0.68$. The pattern across all six: the signal, not the learner, is the bottleneck.

1 Introduction

An agent harness is the scaffold around a model: the loop that decides which tool runs next, which model answers, whether a step is checked, when the episode ends. Inside that loop sit decisions that are made per request, in microseconds, by a rule somebody wrote: "always use the large model", "verify every sub-result", "refine if confidence is low". Each decision has a price in dollars and latency, and the rule does not see it.

Recent work treats the harness itself as the object of optimisation, at design time. The Darwin Gödel Machine evolves self-improving agents [45]; GEPA evolves prompts by reflection [2]; DarwinX [46], Meta-Harness [21], EnvHarness [12] and HarnessX [6] search over harness designs; EvoHarnessBench [16] grows a harness $H_t = \{\text{tools}, \text{skills}, \text{specialist agents}\}$ monotonically by stages and measures what a *fixed* agent policy forgets or gains per axis, reporting that agent-pool expansion alone gives -34.7% backward transfer. SkillOrchestra [41] routes each query to a typed agent in a pool by a learned competence profile with an explicit cost penalty, learned once from traces and then served statically.

A second line trains the orchestrator rather than the harness. MAS-Orchestra [15] generates a complete multi-agent system in one step and optimises the generator with an outcome-only reward; it reports that this holistic formulation outperforms stepwise orchestration, in which an orchestrator chooses one tool or model call at a time [34], and attributes the gap to long-horizon credit assignment under a single terminal reward. Orchestration reward modelling [39] then takes the sub-agents out of the training loop by scoring the plan before it is executed. This report takes the opposite route to the same difficulty. The decisions studied here are per-step and priced, as in the stepwise setting, but credit is assigned per decision rather than per episode: every served decision logs its propensity, the objective is quality minus a learned price of cost, and a candidate policy is valued on logged traffic by doubly-robust estimators with bootstrap intervals before it is promoted. Holistic generation avoids the credit-assignment problem by removing the steps; the controller here keeps the steps and addresses it with the estimators. ToolOrchestra's efficiency term is, to our knowledge, the closest prior use of a priced orchestration reward; it is an episode-level reward without a per-decision estimator or a promotion gate, and the stepwise result reported for it by Ke et al. [15] is a transplanted baseline rather than its native setting, so the comparison between the two formulations under a common price remains open.

This report studies the complementary problem. The harness is frozen; what moves is a small controller at its per-step decision points, trained from logged traffic, priced per request, and put into service only through a statistical gate. Three artefacts make up the method:

- **A priced controller.** A policy over (op, tier) pairs whose objective is quality minus a learned price of cost, trained offline by constrained group-relative policy optimisation on replay (or advantage-weighted regression with inverse-propensity weights), with the Lagrange multiplier found by damped dual ascent against a per-task budget and the served policy the iterate average.
- **An off-policy promotion gate.** Every served decision logs its propensity, exploration included, so a candidate policy can be scored on logged traffic by IPS, SNIPS and doubly-robust estimators with percentile-bootstrap confidence intervals, and is promoted only when the DR lower quality bound and the DR upper cost bound clear two bars.
- **Drift metrics.** Priced backward and forward transfer in the sense of Lopez-Paz and Ranzato [23], applied to the priced objective of the controller as the harness is repriced, re-versioned or grown.

Six experiments exercise these pieces, in increasing realism: a simulator pipeline (gate, drift, retrain); a counterfactual collection on CRMArena-Pro [13]; the gate's sample-size behaviour on that log; a 1.7B-parameter LLM as the controller; process-verified multi-agent orchestration; priced Socratic self-refinement; and the centrepiece, a continually trained orchestrator over a frozen multi-agent structure under scripted harness stages. Several results are negative, and they are reported as results: they say where the signal needed by a learned controller is and is not present, which is the question that decides whether a runtime controller earns its place at all. Every number in this report is taken from the committed result files of the public repository; nothing is re-estimated here.

2 Problem setting

The harness as a frozen decision process. A harness exposes, at each step t of an episode on task x, an observable state s_t and a legal set $\mathcal{A}(s_t)$ of actions $a = (\text{op}, \text{tier})$: an operation the harness can run (answer, retrieve, verify, escalate, delegate, re-plan, call a tool, engage a skill) on a model tier (cheap, mid, frontier). The harness's own transition $s_{t+1} \sim P_H(\cdot \mid s_t, a_t)$ is fixed; the controller $\pi(a \mid s)$ only chooses among what is legal. Each step has a realised cost c_t (tokens $\times$ list price) and latency ℓ_t; a terminal quality $q \in [0,1]$ is the harness's own score. We write $C = \sum_t c_t$ and $L = \sum_t \ell_t$. The harness is a frozen MDP over the controller's decision points, and the controller is the only thing trained.

Priced objective. The product specification is a budget, so the objective is

$$J(\pi) = \mathbb{E}_\pi[q] - \lambda\, \mathbb{E}_\pi[C] - \mu\, \mathbb{E}_\pi[L], \tag{1}$$

with $\mu = 0$ in every experiment here (latency is logged but not priced), and the constrained form

$$\max_\pi \ \mathbb{E}_\pi[q] \quad \text{s.t.} \quad \mathbb{E}_\pi[C] \le B, \qquad \mathcal{L}(\pi, \lambda) = \mathbb{E}_\pi[q] - \lambda\big(\mathbb{E}_\pi[C] - B\big),\ \lambda \ge 0. \tag{2}$$

The price λ is learned by projected dual ascent on an exponential moving average $\hat{c}$ of realised episode cost, with a proportional damping term (the PID-Lagrangian of Stooke et al. [33] without the derivative term; cf. RCPO [37]):

$$\hat{c}_k = \rho\, \hat{c}_{k-1} + (1-\rho)\, \bar{C}_k, \qquad \lambda^{\text{int}}_k = \Big[\lambda^{\text{int}}_{k-1} + \eta\,(\hat{c}_k - B)\Big]_+, \qquad \lambda_k = \Big[\lambda^{\text{int}}_k + k_p\,(\hat{c}_k - B)\Big]_+ . \tag{3}$$

Primal–dual iterates orbit the budget rather than converging to it, so the served policy is the iterate average $\bar{\pi}(a \mid s) = \frac{1}{|\mathcal{S}|}\sum_{W \in \mathcal{S}} \pi_W(a \mid s)$ over weight snapshots $\mathcal{S}$ from the second half of training [25], and $\bar{\lambda}$ is the mean price over the same half.

Policy class and logged propensities. The linear controller is a softmax over soft-clipped logits, mixed with ϵ-exploration:

$$z = L\tanh\big(Wx/L\big), \qquad \pi_\epsilon(a \mid x) = (1-\epsilon)\,\mathrm{softmax}_{\mathcal{A}}(z)_a + \frac{\epsilon}{|\mathcal{A}|}, \tag{4}$$

where $x = \phi(s)$ is a feature vector and the softmax is over the legal set. The logit clip keeps the softmax steerable by λ and keeps importance ratios bounded. The propensity written to the log is $\mu_t = \pi_\epsilon(a_t \mid x_t)$, exploration included, so every served log is usable for off-policy evaluation (the logging discipline of Bottou et al. [4], Li et al. [22]).

Constrained GRPO on replay. When the harness can be re-rolled from a logged state (a simulator or a counterfactual lookup table), training samples a logged state s, draws a group of G continuations from the policy *as served* (π_ϵ, so the cost the dual variable tracks is the served cost), scores each by $R_i = q_i - \lambda C_i$ and uses the group-relative advantage with the group mean as baseline [31]:

$$A_i = R_i - \frac{1}{G}\sum_{j=1}^{G} R_j, \qquad \nabla_W \mathcal{L} = \sum_i A_i \sum_t \nabla_W \log \pi_\epsilon(a_t^i \mid x_t^i), \tag{5}$$

$$\nabla \log \pi_\epsilon(a \mid x) = \frac{(1-\epsilon)\,\pi(a \mid x)}{\pi_\epsilon(a \mid x)}\,\nabla \log \pi(a \mid x). \tag{6}$$

The last factor is the score correction for the ϵ-mixture: the gradient of the mixed policy is the bare softmax's score scaled by the share of the served probability that the softmax contributes.

Advantage-weighted regression with inverse propensities. When no replay exists, the trainer is AWR [27] on logged steps with return $R_i = q_i - \lambda\, C_i^{\rightarrow}$ (cost-to-go from the step), a per-bucket value baseline $\hat{V}$, and weights corrected by the logged propensity and self-normalised over the batch:

$$A_i = R_i - \hat{V}(b(x_i)), \qquad w_i = \frac{\exp(A_i/\beta)/\mu_i}{\sum_j \exp(A_j/\beta)/\mu_j}, \qquad \max_W \sum_i w_i \log \pi(a_i \mid x_i). \tag{7}$$

Here λ is a fixed price. Logged counterfactual arms, when present, are added as extra records (offline RL can only reweight what the log contains).

Off-policy estimators. For an episode i with per-step rewards r_t^i (the terminal quality, or the per-step cost) and cumulative ratio $\rho_{0:t}^i = \prod_{k\le t} \pi(a_k^i \mid s_k^i)/\mu_k^i$ clipped at $\rho_{\max} = 50$,

$$\hat{V}_{\mathrm{IPS}} = \frac{1}{n}\sum_i \sum_t \rho_{0:t}^i\, r_t^i, \qquad \hat{V}_{\mathrm{SNIPS}} = \sum_t \frac{\sum_i \rho_{0:t}^i\, r_t^i}{\sum_i \rho_{0:t}^i}, \tag{8}$$

$$\hat{V}_{\mathrm{DR}} = \frac{1}{n}\sum_i \Big[\hat{V}(s_0^i) + \sum_t \rho_{0:t}^i\big(r_t^i + \hat{V}(s_{t+1}^i) - \hat{Q}(s_t^i, a_t^i)\big)\Big], \qquad \hat{V}(s) = \sum_a \pi(a \mid s)\,\hat{Q}(s,a), \tag{9}$$

the per-decision importance-sampling estimator of Precup et al. [28], the self-normalised estimator of Swaminathan and Joachims [35], and the sequential doubly-robust estimator of Jiang and Li [14] (see also 38). The direct model $\hat{Q}$ is a per-feature-bucket mean fitted from the log and iterated under the candidate, i.e. bucketed fitted-Q evaluation [20]; the clip biases DR only through the direct-model residual. On a first-step contextual bandit (one decision per task, fixed continuation) the bandit form of Dudík et al. [8] applies with a *cross-fitted* direct model:

$$\hat{V}_{\mathrm{DR}} = \frac{1}{n}\sum_i \Big[\sum_a \pi(a \mid x_i)\,\hat{Q}^{(-i)}(x_i, a) + w_i\big(r_i - \hat{Q}^{(-i)}(x_i, a_i)\big)\Big], \qquad w_i = \frac{\pi(a_i \mid x_i)}{\mu_i}, \tag{10}$$

where $\hat{Q}^{(-i)}$ is fitted leaving out task i (the cross-fitting argument of Chernozhukov et al. [7]: an in-sample direct model shares the logged reward it corrects and shrinks the correction toward zero at small n). For multi-step orchestration logs with 2–4 decisions over 12 actions, a per-decision variant treats each logged decision as a bandit sample with single-step ratio $w_t = \pi(a_t \mid x_t)/\mu_t$ and reward the episode's quality or cost-to-go; it values changing one decision while the rest of the episode stays as logged, trading bias for ratios bounded by $|\mathcal{A}|/\epsilon$. Every estimate is reported with its effective sample size $\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ [17].

The gate. Confidence intervals are percentile bootstraps over episodes (over tasks, with all logged arms of a task together, in the bandit case), 200 resamples, refitting the direct model on every resample so the DR interval includes direct-model variance. The decision is

$$\textsc{promote} \iff \hat{q}_{\mathrm{DR}}^{\,\mathrm{lo}} \;\ge\; q_{\mathrm{prev}} - \epsilon_q \quad \text{and} \quad \hat{c}_{\mathrm{DR}}^{\,\mathrm{hi}} \;\le\; B + \mathrm{tol}, \tag{11}$$

with q_{prev} the incumbent's quality: its on-policy log mean if it is the behaviour policy, its own DR estimate on the log otherwise, or the DR quality recorded at the incumbent's own promotion after a repricing (an over-budget incumbent would otherwise set a bar no budget-respecting candidate can pass). Defaults are $\epsilon_q = 0.03$ and tol = \$0.005 (sim units 0.02).

Priced transfer. Let $R_{i,j}$ be the held-out priced value J_j of the policy in service at the end of stage i, evaluated on stage j, with J_j priced at the stage's own $\bar{\lambda}_j$. Following Lopez-Paz and Ranzato [23],

$$\mathrm{BWT}_j = R_{T,j} - R_{j,j} \ (j < T), \qquad \mathrm{FWT}_j = R_{j-1,j} - b_j, \tag{12}$$

with b_j the value of a uniform policy on stage j. For the two-version case, priced-BWT $= J$(frozen on v_2) $- J$(frozen on v_1) and recovery $= J$(served on v_2 after retrain and gate) $- J$(frozen on v_2).

3 Platform

Harness interface. A host harness implements `reset`, `legal_actions`, `step` and `quality`; a simulable one also implements `snapshot` and `restore`, which is what constrained GRPO on replay needs. Its operations are the ops, its model pool the tiers, its version string a feature.

Trace schema. One record per controller decision: features, action, propensity (exploration included), the legal set the behaviour chose from, realised cost and latency, and, when the harness is simulable, a state snapshot for re-rolls. One episode per task carries the terminal quality and metadata (the agent's answer, the queries, the evaluator's calls).

Call log. Every model call is logged with its model, token counts, priced cost, task and purpose, and a hard budget check on each call's worst-case cost runs before the call is sent, so collection is fail-closed and resumable per (task, arm).

Simulator. Tasks have a latent difficulty (easy with probability 0.65, hard 0.35) that the controller never sees; it sees a proxy that is correct 85% of the time. Twelve actions, $\{\textsc{answer}, \textsc{retrieve}, \textsc{verify}, \textsc{escalate}\} \times \{\mathrm{cheap}, \mathrm{mid}, \mathrm{frontier}\}$, each with its own cost, latency and quality tables; at most four steps, terminal ops only on the last. Version v_2 (the drift event) adds CACHED_RETRIEVE, near-free evidence that only works on easy tasks; ages the verifier so that VERIFY adds nothing on hard tasks; and reprices the frontier tier $\times 1.5$. Costs are in simulator units.

CRMArena-Pro host. CRMArena-Pro [13] is the first real host harness. Its ReAct agent [43] runs one closed loop with one fixed model and no per-turn hook, so the harness re-expresses that loop as `step()` and reuses everything else unchanged: task records, prompts, schema rendering, the live CRM connector and the benchmark's own evaluator (exact match, F1 for fuzzy categories, a privacy judge). Actions are `CONTINUE@tier` (one ReAct turn on that tier) and `ANSWER@tier` (force the final answer now); only `ANSWER@*` is legal at the 12-turn cap. Tiers are `claude-haiku-4-5` (cheap), `claude-sonnet-5-5` (mid) and `claude-opus-5-5` (frontier). Features are 43-dimensional: a bias, a one-hot over the 22 task categories, org type, turn one-hot and fraction, tool calls so far, cost so far, whether the last query returned rows, came back empty or errored, and a five-bucket context size. The evaluator's extractor and judge run on the cheap tier and are not charged to the controller's episode cost. Deliberate deviations from the benchmark's run script: no sampling parameters are sent (the mid and frontier models rejected them), calls are bounded in output tokens and time, observations are truncated to 20,000 characters, and a missing import in the benchmark's extractor fallback is supplied.

4 Experiments

4.1 Simulator pipeline: gate, drift, gated retrain

Setup. 3,000 episodes on v_1 under a uniform behaviour policy; constrained GRPO on replay at budget $B = 0.30$ (learned $\bar{\lambda} = 0.358 \pm 0.027$ over seeds); gate the candidate against the behaviour log. Then drift: v_2 plus a scripted reprice(mid, 1.2); 6,000 fresh v_2 episodes served by the frozen policy with $\epsilon = 0.3$; retrain warm-started from the frozen policy; gate the adapted policy against the frozen one with the quality bar taken from the frozen policy's v_1 promotion. Ground truth is Monte Carlo over 20,000 fresh tasks with expected success probability. The whole pipeline is repeated over five replicate seeds (seed s draws every collection, training and bootstrap seed from s; Table 19); numbers are mean ± SE over seeds, and the per-seed values are in Table 25.

Results. Figure 1 shows both gates. On v_1 the learned price is $\bar{\lambda} = 0.358 \pm 0.027$; DR puts the candidate at quality 0.884 ± 0.009 (mean 90% CI $[0.851, 0.910]$) and cost 0.308 ± 0.001 (upper bound 0.311); the decision is PROMOTE in 5 of 5 seeds, and the ground truth (0.878 ± 0.001 at 0.306 ± 0.002) lies inside the DR interval in 4 of 5 (in the fifth it is 0.001 below the lower bound). IPS and SNIPS sit low (0.760 ± 0.022 and 0.797 ± 0.014): the uniform behaviour over 12 actions and up to 4 steps gives cumulative ratios in the hundreds against a near-deterministic candidate, which the clip at 50 truncates. After the drift the frozen policy keeps its quality (0.869 ± 0.001) but its cost rises to 0.413 ± 0.003, over budget, so priced-BWT is -0.047 ± 0.003 (Table 1). The adapted policy is scored at 0.869 ± 0.004 and cost 0.298 ± 0.004 (truth 0.867 ± 0.001 at 0.299 ± 0.004, inside the DR interval in 5 of 5); served, it would recover $+0.039 \pm 0.003$. The gate promotes it in only 2 of 5 seeds. Every revert fails on quality alone, and all three are false alarms: the bar is the DR quality recorded

at the v_1 promotion minus ϵ_q, and in those seeds that estimate was high (0.901, 0.883, 0.903 against truths of 0.877–0.878), so a candidate within 0.01 of the frozen policy's quality and inside the budget could not clear it. The served recovery is therefore $+0.015 \pm 0.009$ (+0.033 and +0.042 where the gate promoted, zero elsewhere). Drift hits cost, not quality: the frozen controller keeps paying for an aged verifier and a repriced frontier, and the retrain restores the budget at equal quality; the quality bar inherited from an optimistic earlier estimate is what keeps it out of service, the same false-alarm mode the continual study finds at larger scale (Section 4.7).

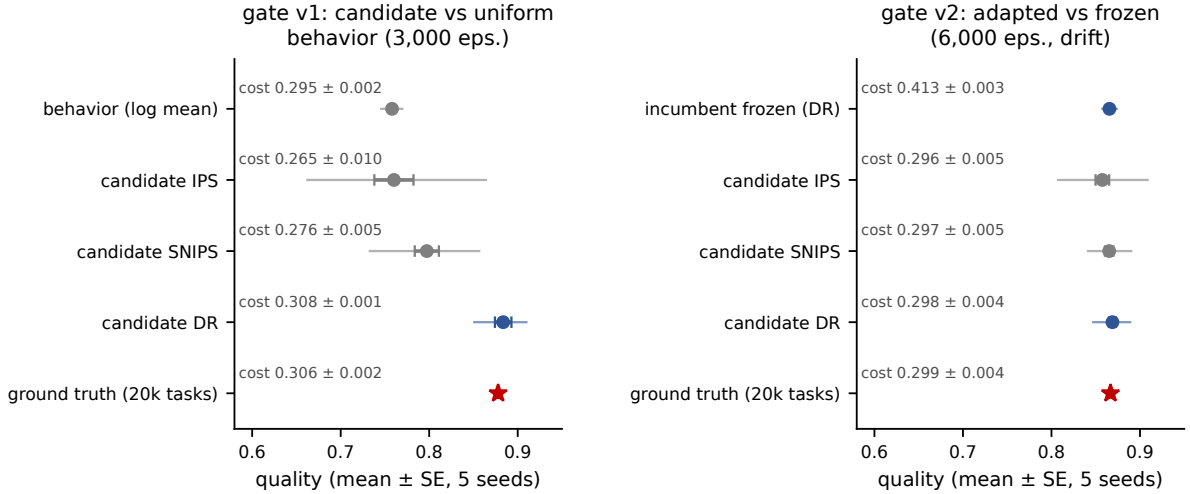


Figure 1: Simulator gates over 5 seeds. Left: the constrained-GRPO candidate scored against 3,000 uniform-behaviour episodes on v_1; right: the adapted policy scored against 6,000 v_2 episodes served by the frozen policy. Points are means over seeds of each estimate with SE bars; thin lines are the mean 90% bootstrap CI; the star is Monte Carlo ground truth (for v_2, the adapted policy whether or not promoted). The gate decides on DR (blue). From `results/sim_seeds/`.

Table 1: Simulator drift table, ground truth, mean ± SE over 5 seeds (J at each seed's own $\bar{\lambda}$). priced-BWT $= J(\text{ii}) - J(\text{i}) = -0.047 \pm 0.003$; recovery $= J(\text{iii}) - J(\text{ii}) = +0.015 \pm 0.009$ as served after the gate, $+0.039 \pm 0.003$ for the adapted policy.

condition	quality	cost	within budget	J
(i) frozen on v_1	0.878 ± 0.001	0.306 ± 0.002	5/5	0.768 ± 0.008
(ii) frozen on v_2	0.869 ± 0.001	0.413 ± 0.003	0/5	0.721 ± 0.011
(iii-a) adapted on v_2 (whether or not promoted)	0.867 ± 0.001	0.299 ± 0.004	5/5	0.760 ± 0.008
(iii) served on v_2 after the gate (2/5 promoted)	0.868 ± 0.001	0.369 ± 0.028	2/5	0.737 ± 0.014

Limitations. The simulator's difficulty structure, success tables and drift event are hand-written; the run demonstrates the mechanics, not effect sizes. Five seeds are enough to show that the second gate's decision depends on the seed, not to estimate its promotion rate. The sim gate fits its direct model in-sample (buckets hold hundreds of transitions).

4.2 CRMArena-Pro counterfactual collection

Setup. 50 b2b tasks (2–3 per category, all 22 categories), each run on all three tiers with a fixed continuation (`CONTINUE` on that tier until the agent answers, `ANSWER` forced at the cap), so each task's first-step tier is a contextual bandit with a fully observed reward table and the held-out value of any tier policy is an exact lookup. The log holds 150 episodes from 832 model calls; episodes average 5.0 turns and 16 of 150 hit the 12-turn cap. The behaviour is uniform over tiers (propensity 1/3 at step 0, 1 after). The learned decision is the first-step tier from the 43-dimensional step-0 features. Policies are trained under 5-fold category-stratified CV over task ids (train on 40, evaluate on the held-out 10, seed 0), at fixed $\lambda \in \{2, 5, 10\}$ per dollar with AWR+IPS, and under a budget $B = \$0.10$ with constrained GRPO on the lookup table (learned λ).

Results. Table 2 and Figure 2. No learned linear rule clearly beats the best fixed arm held-out: the served AWR policy has lower J than the best fixed arm at every λ (0.350 vs mid 0.368; 0.045 vs cheap 0.104; −0.414 vs cheap −0.236), and its greedy version and the per-category rule beat the best arm by at most 0.08, inside the fold SDs. The features memorise: AWR's J at $\lambda = 2$ is 0.475 in-sample against 0.350 held-out; the constrained policy meets the budget in-sample (0.669 at \$0.099) and overspends held-out (0.553 at \$0.131), where it is dominated by the uniform behaviour (0.571 at \$0.117). Even the context-free arm mixture tuned to B on 40 tasks costs \$0.115 on the held-out 10. The per-task oracle reaches 0.769 at \$0.092, beating every arm on both quality and cost: the headroom is per task, not per category, and the missing piece is an

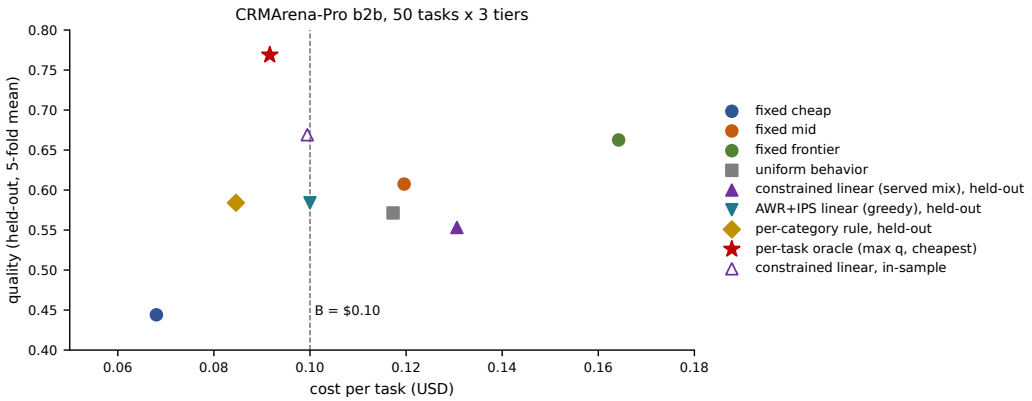


Figure 2: CRMArena-Pro b2b: held-out quality against cost per task (5-fold means, exact lookups). The per-task oracle beats every fixed arm on both axes; no learned policy does. The open triangle is the constrained policy in-sample (0.669 at \$0.099), the filled one held-out (0.553 at \$0.131).

Table 2: Held-out first-step tier policies on CRMArena-Pro b2b (5-fold means; cost in USD per task; across-fold SD of J in brackets, 0.05–0.3 throughout, as large as the gaps between rows).

policy (held-out)	$\lambda = 2$: J	$\lambda = 5$: J	$\lambda = 10$: J	$B =$ \$0.10: quality / cost
AWR+IPS linear, served (stochastic)	0.350 [0.047]	0.045 [0.126]	−0.414 [0.246]	–
AWR+IPS linear, greedy	0.384 [0.076]	0.186 [0.126]	−0.273 [0.272]	–
per-category plug-in rule	0.415 [0.066]	0.161 [0.104]	−0.214 [0.154]	–
constrained cgrpo-lookup (iterate mix)	–	–	–	0.553 / 0.131 (over)
uniform behaviour	0.337 [0.066]	−0.015 [0.164]	−0.601 [0.340]	0.571 / 0.117
fixed cheap	0.308 [0.179]	0.104 [0.227]	−0.236 [0.308]	0.444 / 0.068
fixed mid	0.368 [0.092]	0.010 [0.124]	−0.588 [0.298]	0.608 / 0.120
fixed frontier	0.334 [0.133]	−0.159 [0.317]	−0.980 [0.633]	0.663 / 0.164
per-task oracle	0.639 [0.056]	0.452 [0.080]	0.148 [0.168]	0.769 / 0.092

observable that predicts which tier a task needs. Repricing events (frontier ×1.5, cheap ×3), with costs recomputed from logged tokens, push the frozen all-50 policy ($\bar{\lambda} = 1.36$/USD) over budget by +\$0.029 and +\$0.035: priced-BWT −0.040 and −0.048, in-sample recovery +0.069 and +0.091; but held-out the retrained policies still overspend (about \$0.125) and the gate on the repriced realised log REVERTs all three retrains, which the held-out table says is the right call (recovery −0.012 for frontier ×1.5).

Limitations. Fifty tasks, one org, one seed; one task moves a fold's quality by 0.1. Only the first-step tier is learned, because the continuation was fixed at collection. The log holds one harness version, so version drift cannot be measured on it. Prompt caching was not used, so the roughly 5k-token system prompt is paid on every turn.

4.3 The gate's sample-size result

Setup. The candidate is the constrained policy trained on all 50 tasks (exact value: quality 0.665, cost \$0.0987, within $B + \text{tol} =$ \$0.105). The realised behaviour log is one uniformly drawn arm per task (50 episodes). The gate (cross-fitted DR, $\epsilon_q = 0.03$) is run once, then over 200 independent redraws of the realised log with a 200-resample bootstrap in each.

Results. On the single realised log DR estimates 0.740 $[0.627, 0.855]$ and cost \$0.0960 $[0.0750, 0.1200]$; the quality criterion passes and the cost criterion fails ($0.1200 > 0.105$): REVERT. On the full counterfactual log all three estimators reduce to the exact value (the weights over a task's three arms sum to 3) and the CI is pure task-sampling uncertainty from 50 tasks; the cost upper bound is still \$0.1122 and the decision is still REVERT. Over the 200 redraws (Table 3, Figure 3) DR halves IPS's quality RMSE (0.048 vs 0.094) with 99% coverage of its 90% CI, but its cost CI is \$0.046 wide on average, so the candidate, whose exact cost is \$0.0013 under $B + \text{tol}$, is PROMOTEd in only $12/200 = 6\%$ of redraws. The gate errs on the safe side; buying power means more tasks or a wider tolerance. As a back-of-envelope figure (not a measurement), with CI width scaling as $1/\sqrt{n}$, certifying a \$0.005 margin needs a CI near \$0.01, i.e. roughly 20× the tasks, about 1,000.

Limitations. The in-sample direct model covers 96%/94% with a −0.006 quality bias, a difference inside the noise at $n = 50$; the redraw study reports bias and coverage empirically rather than assuming them.

Table 3: Estimators over 200 redraws of the realised CRMArena-Pro log ($n = 50$, one arm per task). Cross-fitted direct model. PROMOTE in 12/200 redraws (6%).

metric	estimator	bias	RMSE	mean 90% CI width	CI covers exact
quality	IPS	+0.011	0.094	0.338	94%
quality	SNIPS	+0.001	0.048	0.238	100%
quality	DR	+0.004	0.048	0.256	99%
cost	IPS	+0.0001	0.0114	0.0414	92%
cost	SNIPS	−0.0009	0.0102	0.0381	92%
cost	DR	−0.0018	0.0114	0.0461	92%

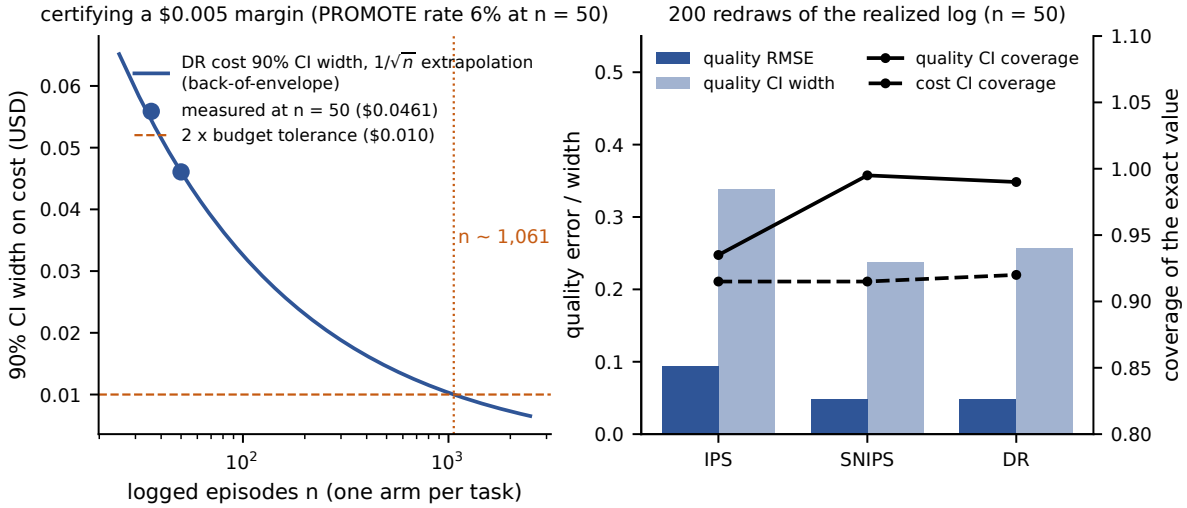


Figure 3: Gate power. Left: the measured DR cost CI width at $n = 50$ and its $1/\sqrt{n}$ extrapolation (a back-of-envelope, not a measurement) against twice the budget tolerance. Right: estimator RMSE, CI width and coverage over the 200 redraws.

4.4 LLM controller: GRPO with LoRA on a 1.7B model

Setup. The controller is Qwen3-1.7B [30] with a LoRA adapter [11] ($r = 16$ on the attention and MLP projections, bf16), trained with GRPO. The agent models stay frozen; the tiers are the arms. The prompt shows the difficulty proxy, the state, per-tier prices, the budget and the legal actions, and the completion must be exactly one tag `<action>TIER OP</action>`; anything else is invalid (reward −1, cost booked as the cheapest safe action). After the first action a fixed continuation answers on the same tier; the reward is $q - \lambda C$ with λ learned by the same damped dual ascent, once per step from the batch mean cost. Each step generates 64 completions (8 prompts × 8 generations in phases 1 and 1b; 4 × 16 in 1c), length 16, lr 10^{-5}, over 2,000 seeded sim tasks with the legal-action order shuffled per task (with a fixed order the base model only ever sampled one or two arms). Three runs on the simulator, each run once on one A100-40GB: `phase1` (λ lr 0.1, k_p 2.0, β_{KL} 0.02, 300 steps); `phase1b` (slower dual step: lr 0.02, k_p 0.5, EMA 0.95; β_{KL} 0.05; entropy 0.01); `phase1c` (150 SFT steps on per-difficulty oracle labels at a price shown in the prompt and drawn from $\{0.0, 0.1, \ldots, 2.0\}$, adapter merged so the KL reference is the SFT policy, then 200 GRPO steps with 16 generations, temperature 1.2, $\lambda_0 = 0.5$). A fourth run (`crm_phase2`) applies the same recipe to CRMArena-Pro fold 0 in lookup mode. Per-run settings are in Appendix A.

Table 4: LLM controller on sim v_1 (2,000 fresh tasks, seed 777; budget $B = 0.30$; the policy's first action then `ANSWER` on the same tier). "Within" means cost $\leq B + 0.02$.

policy	quality	cost	within	invalid
Qwen3-1.7B zero-shot (greedy, no adapter)	0.586	0.048	yes	18.9%
`phase1`, steps 100 / 200 / 300	0.748–0.749	0.155	yes	0%
`phase1b`, step 100	0.885	0.603	no	0%
`phase1b`, step 300	0.832	0.434	no	0.1%
`phase1c`: SFT only, and GRPO steps 100 and 200 (all identical)	0.848	0.330	no	0%
linear controller (`policy_v1_sim.json`), first action only	0.573	0.035	yes	0%
fixed `ANSWER` on cheap / mid / frontier	0.522 / 0.688 / 0.820	0.020 / 0.100 / 0.400	yes / yes / no	0%
best deterministic per-proxy rule (easy: mid ESCALATE; hard: frontier ANSWER)	0.814	0.250	yes	–
best randomised policy at cost exactly B (LP)	0.835	0.300	yes	–
derived: `phase1c` served with the price drawn as in training's last 100 steps	0.837	0.305	yes	–

Results. Table 4 and Figure 4. GRPO alone never found the budgeted optimum. *Group degeneracy*: within a prompt group the completions are almost always identical, 83.5% of groups on average in `phase1b`, which gives zero GRPO advantage, so the only learning signal is λ moving the mean reward between steps and the policy jumps between near-

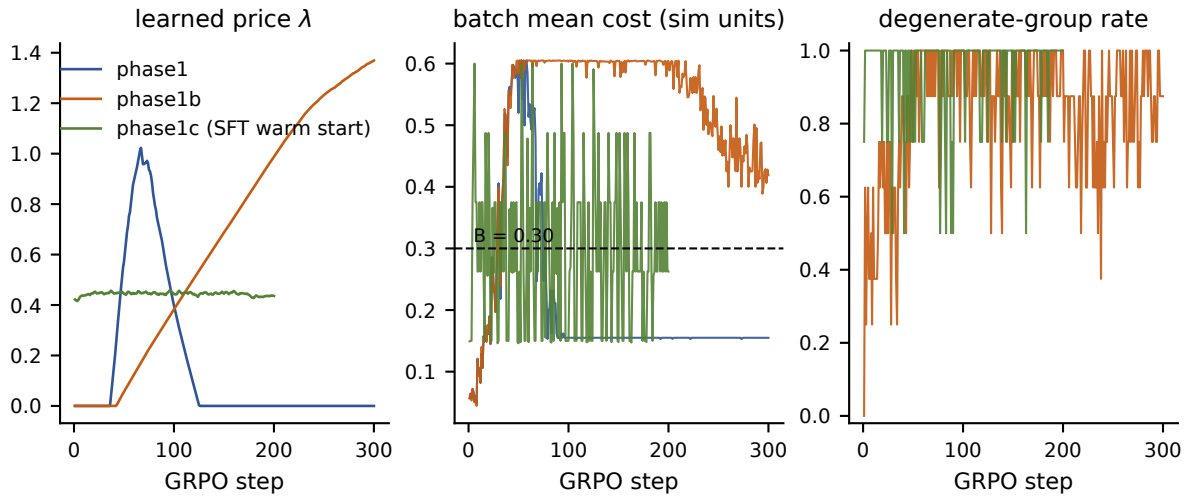


Figure 4: GRPO training logs. `phase1`'s price orbits once and decays to 0 as the policy collapses onto `mid RETRIEVE`; `phase1b`'s price is still rising at step 300 while cost stays above budget; `phase1c`'s price settles at $\lambda \approx 0.44$ with the batch cost alternating between two deterministic rules. Degenerate groups (all completions identical, hence zero advantage) dominate every run.

deterministic arms instead of learning the per-proxy split. *Primal–dual orbit*: in `phase1` the policy swings to `frontier RETRIEVE` (cost 0.60), λ climbs to 1.02 by step 67, the policy collapses onto `mid RETRIEVE` (0.155) and λ decays to 0 with nothing to pull it back; in `phase1b` λ is still rising at step 300 (1.37) and the policy has not reached the budget. Mixing is not the bottleneck: a deterministic per-proxy rule reaches 0.814 at 0.250 and randomisation adds only 0.021; what the runs failed to do is condition on the proxy. In `phase1c` the SFT stage reached 100% held-out agreement with the oracle labels (1,000 prompts, 0% invalid) and splits by difficulty where the oracle does; GRPO then left the policy unchanged on all 2,000 eval tasks (the degenerate-group rate averaged 93% over steps 1–100 and 97% over 101–200 even after the rollout temperature was raised from 1.2 to 1.5 at step 20). GRPO's contribution is the price: λ settles at 0.44 (mean 0.443 over the last 100 steps), which is the slope at which the hard-proxy action flips. At that price the SFT policy ($J = 0.702$) ties the oracle rule (0.703) and the LP (0.702). No single greedy policy meets $B = 0.30$: at any one shown price the policy is a deterministic rule costing 0.33 or 0.15; training met the budget by alternating the shown price over time (batch cost 0.301 and 0.296 over the two halves), and serving can reach it by drawing the shown price (the price-conditioned analogue of iterate averaging): 0.837 at 0.305, derived from the two greedy rules rather than decoded end to end. On CRMArena-Pro fold 0, the SFT policy ignores the shown price (the same per-category rule at every λ, about \$0.075, under budget), so λ fell to 0 by step 17; GRPO moved training-set quality from 0.705 to 0.730 and left the 10 held-out tasks unchanged at 0.700 / \$0.073, against fixed frontier 0.700 / \$0.142 and the fold-0 linear policy 0.590 / \$0.143. That is one 10-task fold where one task is 0.1 of quality and the 5-fold mean of the same category rule is 0.584 / \$0.085 (Table 2): a single draw, not evidence of generalisation or of the LLM beating the linear controller.

Limitations. Each GPU run ran once; no seed variance. The phase-1 table truncates the linear policy's continuation (its first action is `cheap RETRIEVE` because it was trained to keep going; with its own continuation it scored 0.876 at 0.301 in the run that produced that file), so its row is not its real performance. Showing λ rounded to 0.1 puts $\bar{\lambda} = 0.443$ on the over-budget side of the flip point.

4.5 Process-verified multi-agent orchestration

Setup. A planner (cheap tier) decomposes a CRM request into 1–3 sub-queries; a data worker (mid) runs one sub-query at a time through the harness's own step path; a verifier (cheap) checks a sub-result against its goal (a structural pre-check fails at \$0 on query errors); the answer is the harness's forced `ANSWER@mid`. After every worker step an orchestration policy chooses $\{\text{VERIFY}, \text{SKIP_VERIFY}, \text{REPLAN}, \text{ANSWER}\} \times$ tier; a fail retries the sub-query once; at most 8 role calls per episode. Fixed modes `always_verify` and `never_verify`, plus a uniform exploration mode with logged propensities, were run on the same 50 b2b tasks as the counterfactual log: 130 episodes (50 + 50 + 30). An orchestration reward model (ORM: logistic regression and a 16-unit MLP over process features) is evaluated out of fold over tasks.

Results. The headline is negative (Table 5): the decomposition scores below the single mid-tier agent (0.480 vs 0.608) at lower cost and is dominated on the priced frontier at both prices (by mid at $\lambda = 2$, by cheap at $\lambda = 5$); paired on the 50 tasks the single mid agent wins 11 and loses 5. This is the compute-matched single-agent comparison MAS-Orchestra [15] reports as the bar, and it is not cleared. Three mechanisms are visible in the log: the plan is fixed up front by a cheap model with no data in hand while the single agent picks each query after seeing the last result; `never_verify` gives the strong model about 2.7 worker turns and a forced answer against the single agent's about 5; and under `always_verify` 47

Table 5: Decomposition against the single ReAct agent on the same 50 CRMArena-Pro b2b tasks (single-agent rows are exact lookups of the counterfactual log). The six confidentiality tasks score 0 for every system, so the ceiling is 0.88.

system	accuracy / quality	cost / task	$J(\lambda = 2)$	$J(\lambda = 5)$
single agent, fixed cheap	0.444	\$0.0680	0.308	0.104
single agent, fixed mid	0.608	\$0.1196	0.368	0.010
single agent, fixed frontier	0.663	\$0.1642	0.334	−0.159
single agent, per-task oracle (max quality, cheapest)	0.769	\$0.0916	0.586	0.311
MAS `always_verify`	0.480	\$0.0906	0.299	0.027
MAS `never_verify`	0.461	\$0.0821	0.297	0.051
MAS `controller:uniform` (30 tasks, exploration)	0.535	\$0.0758	0.384	0.156

Paired, same 50 tasks: `always_verify` vs `never_verify`: 6 wrong→right, 5 right→wrong, net +1 ($\Delta q = +0.019 \pm 0.067$ SE, +\$0.0086). `always_verify` vs single mid: both right 19, MAS right / mid wrong 5, MAS wrong / mid right 11, both wrong 15 (net −6).

Table 6: Orchestration reward model, out of fold over tasks (130 episodes, base rate $P(\text{correct}) = 0.485$, base-rate Brier 0.250).

model	features	AUROC	Brier	ECE (5 bins)
logreg	process features (default)	0.469	0.281	0.114
logreg	process − verifier features	0.451	0.280	0.140
logreg	process + category one-hot	0.831	0.165	0.040
mlp	process features (default)	0.456	0.263	0.053
mlp	process + category one-hot	0.831	0.164	0.035
–	category alone, no process feature	0.821	0.166	–

of 50 episodes hit the 8-call cap because 65% of verifier calls fail and each fail spends a retry. Verification bought +1 task for +10% cost, and the paired table says that is noise (6 flips each way, all through a fail and a retry). The verifier is an alarm, not a judge: it flags 24 of 26 wrong episodes ($P(\text{flag} \mid \text{wrong}) = 0.92$, silent failures 2/50) and 23 of 24 correct ones ($P(\text{flag} \mid \text{correct}) = 0.96$), so $P(\text{wrong} \mid \text{flagged}) = 0.51$ against a base rate of 0.52. The ORM (Table 6) finds that process features carry no out-of-fold signal (AUROC 0.469, Brier above the base rate) and the task category carries most of it (0.821 alone; 0.831 with process features), with CV over tasks so that this is category-level difficulty transferring to unseen tasks, resting on 2–3 training tasks per category. The priced-controller hook was exercised, not evaluated: AWR on the 30-episode ORM-relabelled exploration log changes nothing detectable per decision (per-decision SNIPS 0.563 $[0.373, 0.701]$ against the behaviour's 0.567, ESS 55 of 64), and the episode-level estimate has ESS 7 of 30.

Limitations. The verifier is a cheap model of the same family checking a mid model; no human process label. The 8-call cap interacts with a high-fail verifier, so a higher cap or a better verifier is a different system. Sequential sub-queries in one growing context.

4.6 Priced Socratic self-refinement

Setup. One Socratic self-refinement round [32] on a CRMArena-Pro answer: a cheap probe decomposes the answer into 1–3 claims with self-reported confidences (about \$0.003, so a gate can look before it pays); each claim is re-solved by $k = 2$ worker mini-episodes on the chosen tier (≤ 3 turns, the SOQL runs on the real org); a rewrite on that tier produces the refined answer; the benchmark's evaluator scores it. The initial answers are the cheap-tier episodes of the counterfactual log (already scored and paid). The run logged a tree, a refinement on every tier for every task, so every policy over $\{\text{STOP}, \text{REFINE@cheap}, \text{REFINE@mid}\}$ is an exact lookup: `never`, `always_1:t`, `uncertainty_gated:`τ`:t` (τ swept 0.3–0.9 on the minimum claim confidence), and a softmax-linear priced controller over 36 features trained by exact policy gradient on the tree (budget $B = \$0.10$ with learned λ, or fixed $\lambda \in \{2, 5\}$; 5-fold CV over tasks). Of 42 tasks probed, 41 completed on both tiers. One task is 0.024 of quality.

Table 7: Refinement harm after one round (41 tasks). "fix" is wrong→right out of initially wrong, "harm" right→wrong out of initially correct.

tier	changed	fix	harm	mean Δq	round cost	worker turns
cheap	85% (35/41)	13% (3/23)	28% (5/18)	−0.049	\$0.0962	14.7
mid	44% (18/41)	13% (3/23)	6% (1/18)	+0.049	\$0.2380	13.9

Results. The probe's confidence is weakly informative: AUROC 0.65 (90% CI 0.51–0.79, $n = 41$) of 1 − min claim confidence for predicting a wrong initial answer; the decomposer is pessimistic about everything (minimum claim confidence never above 0.75, below 0.4 on 26 of 41 tasks whether right or wrong), and the re-solve signal (AUROC 0.67 cheap, 0.55 mid) is no better than the free one; 22% of mid worker samples are "None". Refinement harms as often as it helps (Table 7): a cheap round fixes 3 of 23 and breaks 5 of 18 ($\Delta q = -0.049 \pm 0.069$), a mid round fixes 3 and breaks 1 ($+0.049 \pm 0.049$), both within one SE of zero. Policies (Figure 5): `never` 0.440 ± 0.078 at \$0.068; `always_1:cheap` 0.391 at \$0.167; `always_1:mid`

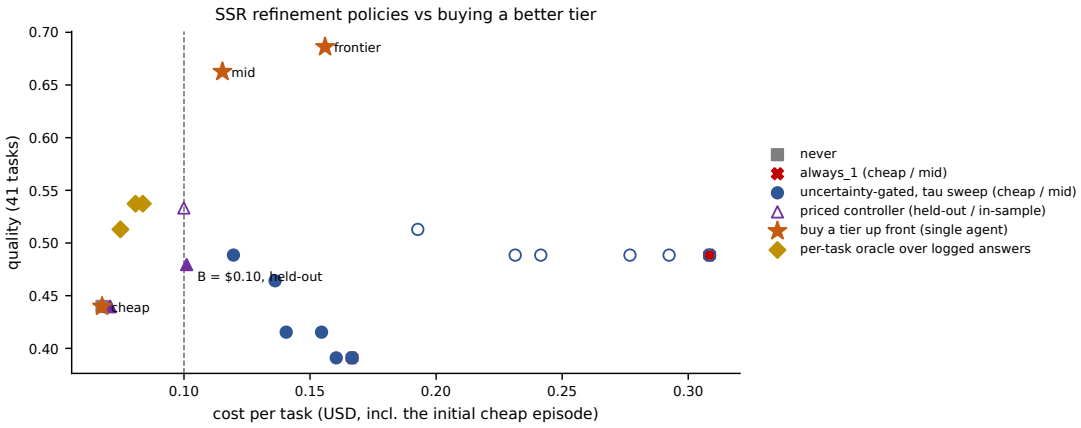


Figure 5: SSR policies on the logged tree (41 tasks, cost includes the initial cheap episode), from `frontier.csv`. Open circles are mid-tier gates, filled are cheap-tier. The single-agent mid arm at $0.115 dominates every refinement policy.

0.489 at $0.309; the best fixed cheap gate ($\tau = 0.3$) 0.489 at $0.120; the budgeted controller 0.480 ± 0.077 at $0.101 held-out (paired $+0.040 \pm 0.033$ over `never`, refining 20% of tasks, $\bar{\lambda} = 0.09$/USD because it sits under budget; 0.533 at $0.100 in-sample), and the fixed-λ controllers learn `never`. Even the clairvoyant choice among the three logged answers per task reaches only 0.537 at $0.084. H1 (refine where confidence is low) is not established; H2 (always-refine is dominated under a budget) is supported, on the cheap tier by `never` itself; H3 (a priced gate reaches the frontier) only in the weak sense that the held-out controller sits on the frontier by learning to refine rarely. The comparison that matters is refine-the-cheap-answer against buy-mid-up-front, and buying the tier wins by a wide margin: the single-agent mid arm on the same 41 tasks scores 0.663 ± 0.074 at $0.115; the best cheap-refinement policy at that price is -0.174 ± 0.085 paired for +$0.004; `always_1:cheap` is -0.272 ± 0.099; even refining on mid at 2.7× the cost is -0.174 ± 0.092; frontier up front (0.686 at $0.156) beats every refinement policy by 0.17–0.30. The sub-questions need the same schema reasoning that made the cheap agent fail, and the workers inherit it.

Limitations. One round, 41 tasks, one initial arm; the τ sweep is tuned on the tasks it is evaluated on. The in-sample controller row shows what 22 category one-hots memorise on 41 tasks. Agreement with $k = 2$ is weak, and no sampling parameters are sent.

4.7 The continual orchestrator

Harness definition. The multi-agent structure of Section 4.5 is expressed as a typed harness in the terms of SkillOrchestra and EvoHarnessBench: an agent pool of typed specialists (agent id, tier, mode, stage) with modes planner, data-worker, verifier and answer; tools the orchestrator can `call@`; skills it can `engage@`; and a competence handbook over the 22 task categories (the axis the real run found carries the signal). The harness $H_t = \{\text{tools}, \text{skills}, \text{agents}\}$ grows monotonically by stages, $H_1 \subseteq H_2 \subseteq \cdots$, plus a price axis; the structure at each stage is frozen and the *policy* at the decision point is what adapts, so the question is EvoHarnessBench's turned around: when H expands under an orchestrator that keeps learning, what happens to the policy's backward and forward transfer, and does the gate keep it safe without making it too slow to matter? Eighteen actions: `delegate@data-1..4`, `verify@verify-1,2`, `replan@plan-1..3`, `answer@answer-1..4`, `call@schema_lookup`, `call@aggregate_calc`, `engage@soql_aggregate`, `engage@date_window`, `engage@decompose_fine`; and 88 features (the real orchestrator's 66, computed by its own feature function on a view of the simulated state, plus availability flags, tools called, skills engaged, stage and a last-fail flag). Unavailable capabilities have fixed zero feature slots and zero logits, so a policy trained at stage g keeps its weights when stage $g + 1$ arrives.

Simulator and calibration. Days of traffic under several harness changes are beyond what the real collection covers, so the logged process is re-expressed as a seeded simulator whose decision points, legal-set rules and feature layout are the real orchestrator's. Every parameter is either measured from the two committed logs (130 MAS episodes, 150 single-agent episodes), with its count, or marked assumed (Appendix D). Measured: plan-length distribution (4/22/104 of 130), worker step kinds per tier, the retry re-draw, the verifier's per-call fail rates by step kind (72 of 124 on rows, 23 of 24 on empty, 12 of 12 structural at $0) and its episode-level $P(\text{flag} \mid \text{wrong}) = 0.92$ and $P(\text{flag} \mid \text{correct}) = 0.96$, token means per role and tier, per-category accuracy, and the configuration. Assumed: whether a rows step is the *right* rows ($P(\text{good} \mid \text{rows}) = 0.60/0.75/0.85$ by tier), the logit slope from plan coverage to the answer (1.5, centred so `never_verify` reproduces the category base rates), a task-level logit noise (sd 0.4), and everything that arrives after stage 0. On the stage-0 harness the two fixed modes reproduce the real run's process numbers (Table 8).

Table 8: Simulator calibration check: stage-0 harness, 1,000 simulated tasks per mode against the real run's 50 tasks per mode (sim / measured).

mode	accuracy	cost/task	role calls	verifier calls	cap hits	P(flag | wrong)	P(flag | correct)
`always_verify`	0.446 / 0.480	\$0.0894 / \$0.0906	7.87 / 7.88	2.94 / 2.94	95% / 94%	0.95 / 0.92	0.93 / 0.96
`never_verify`	0.459 / 0.460	\$0.0827 / \$0.0821	4.75 / 4.74	0 / 0	0% / 0%	–	–

Pre-registered stages. Both schedules were written down before the first full run and not changed afterwards (Table 9). The `mild` schedule keeps the days and axes with smaller effects: reprice mid $\times 0.8$ / frontier $\times 1.2$, `verify-2` at P(flag | correct) = 0.60, `data-4` at rows 0.78 / good 0.75, `answer-4` mid-grade on 3 categories, half the tool and skill bonuses, planner v2 at 1–3 sub-queries.

Table 9: The pre-registered default schedule: 40 days, events on days 9 / 17 / 25 / 33.

stage	day	axis	what changed
g0 baseline	1	–	the real run's pool: plan-1, data-1..3, verify-1, answer-1..3; list prices
g1 reprice	9	price	mid $\times 0.5$, frontier $\times 1.5$
g2 agents	17	agents	+verify-2 (a better verifier: P(flag | correct) 0.96 $\rightarrow$ 0.30 at the same recall), +data-4 (database specialist, cheap), +answer-4 (reporting specialist, cheap)
g3 tools	25	tools	+schema_lookup (the next worker step cannot error), +aggregate_calc
g4 skills	33	skills	+soql_aggregate, date_window, decompose_fine; the planner now decomposes into 1–4 sub-queries

The day loop and comparators. Each day is 96 seeded tasks, the same for every arm. The continual arm serves its incumbent with $\epsilon = 0.1$ and logs propensities; a learner trains every day by constrained GRPO on replay from the last 5 days of logged states (60 iterations, batch 16, group 6; λ learned against B = \$0.090 per task; iterate averaging), warm-started from itself; the gate (sequential DR with 200 bootstrap resamples, $\epsilon_q = 0.05$, tol = \$0.005) scores the learner against the incumbent on the log served since the last harness event (up to 5 days), with the budget net of the policy-independent prefix (planner plus first worker call, \$0.022 at list prices). An AWR+IPS variant (fixed $\lambda = 2$, 200 epochs) runs the same loop. Baselines on the same traffic: `frozen` (the learner as of day 5, served without ϵ, restricted to stage-0 actions); `so_static`, a SkillOrchestra-style router whose handbook is the literal rule competence$[(a, \text{category})] - \lambda_c \cdot \overline{\text{cost}}[a]$ with cost tie-break, learned from the warm-up exploration traces (days 1–4) with λ_c chosen on day 5 by episode-level SN-IPS, never updated (new capabilities scored at the handbook's prior); `so_relearn`, the same router re-fitted after every stage (3 days served with $\epsilon = 0.3$, then a refit and re-tune, the lag counted); `always_verify`; `never_verify`; and a per-stage `oracle` (constrained GRPO on 200 fresh exploration episodes of that stage, 300 iterations, no gate: a reference, not a proven optimum). Held-out: 150 fixed tasks per stage, simulator expected quality, never seen by the gate; $J_g = \text{quality} - \bar{\lambda}_g \cdot \text{cost}$ with $\bar{\lambda}_g$ the stage oracle's learned price (1.25 / 0.85 / 0.81 / 0.66 / 0.73 per USD for g0–g4), at which the budgeted optimum maximises the Lagrangian so J_g has a ceiling and regret is well defined. Metrics follow Eq. (12); in addition, hit_j is the event-day policy's J on stage j minus on stage $j-1$, $\text{recovery}_j = R_{j,j} - R_{j-1,j}$, days-to-recover is the first day of the stage within 0.02 of the oracle, stationary J the mean over the stage's last four days, and cumulative regret the sum over days 6–40 of the oracle's J minus the serving policy's on the day's stage. One seed, on CPU.

Results, stated in order of importance. *(1) The fixed mode `never_verify` wins; against the learned alternatives the continual orchestrator wins on every metric.* Cumulative regret over days 6–40 (Table 11, Figure 6): `never_verify` 0.56, continual (cgrpo, gated) 1.26, `always_verify` 1.99, frozen 2.60, router re-learned per stage 3.03, static router 3.46, continual AWR 3.90. `never_verify`, the all-mid fixed mode, is at the oracle on g1 (0.466 vs 0.460) and within 0.02–0.04 on g2–g4 without learning anything: after the mid repricing, running every sub-query and the answer on the half-price mid tier is the budgeted optimum or near it in this simulator. The continual arm ends every stage within 0.001–0.014 of the oracle and is the only learned arm that recovers at all (days to recover 6 / 6 / 1 / 1); the frozen controller, the static router and the re-learned router never come within 0.02 of the oracle on any stage. The continual arm beats `never_verify` only in the stationary periods of the capability-adding stages (g2 0.429 vs 0.423, g3 0.453 vs 0.434, g4 0.448 vs 0.428) and loses that back during the recovery days. Over 5 seeds of the same configuration (Section 4.7, variant A; mean ± SE), the continual-minus-`never_verify` stationary J is -0.050 ± 0.011 (g0), -0.009 ± 0.002 (g1), -0.014 ± 0.007 (g2), $+0.011 \pm 0.010$ (g3) and $+0.031 \pm 0.007$ (g4), and the whole-stream regret is continual 1.06 ± 0.21, `never_verify` 0.74 ± 0.08, `always_verify` 1.97 ± 0.09: the fixed mode's whole-stream win and the learner's margin at the last stage are both outside seed noise; the g2–g3 margins are not. The honest answer to "does the learner beat the alternatives" is: it beats every design-time learner, and the thing that beats it is knowing the right fixed policy in advance, which the growing pool makes less likely with every stage (the fixed mode's gap to the oracle grows from -0.006 at g1 to -0.038 at g4).

(2) The whole-stream headline is a cold-start artefact: the learner crosses `never_verify` around day 22–25 and wins every stage window after that, while `never_verify` falls further below the oracle at each capability stage. Figure 7. Served

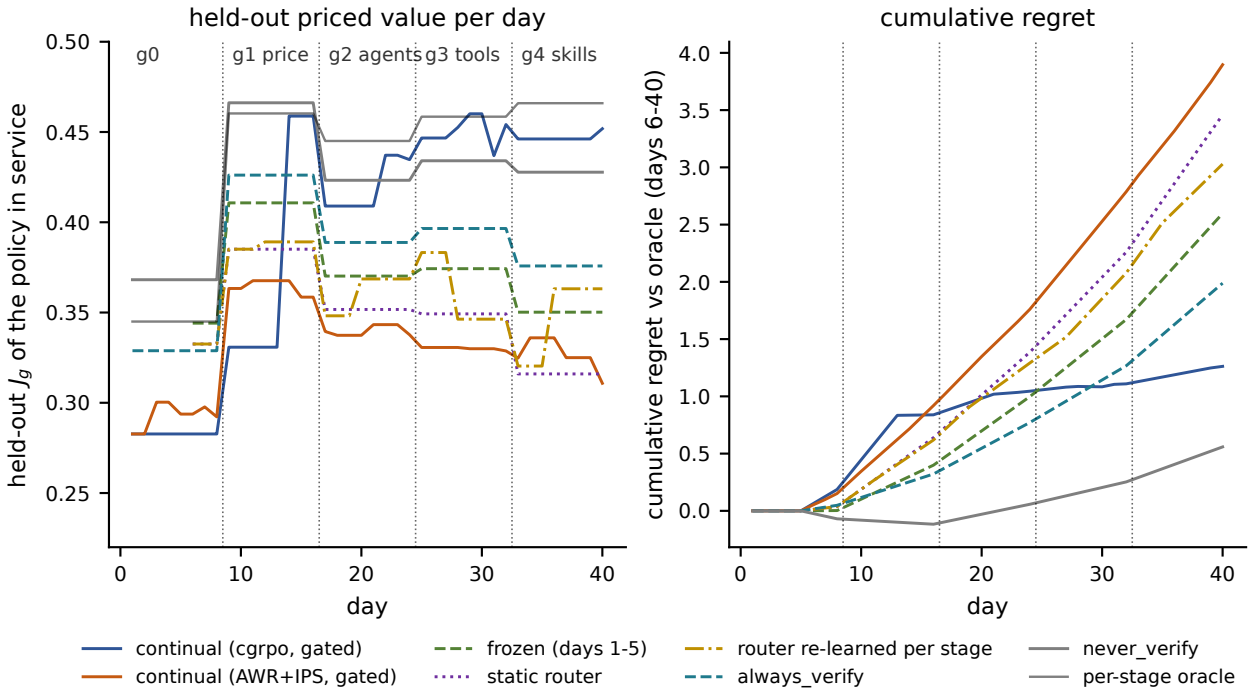


Figure 6: The continual orchestrator over 40 days (default schedule, from `days.jsonl`). Left: held-out J_g of the policy in service each day; the grey upper line is the per-stage oracle. Right: cumulative regret to the oracle from day 6. The continual (cgrpo) arm is the only learned arm that recovers after every stage; `never_verify` has the lowest regret of all.

Table 10: Per-event transfer on the default schedule (held-out J of the policy in service; oracle J per stage g1–g4: 0.460 / 0.445 / 0.458 / 0.466; uniform 0.351 / 0.324 / 0.317 / 0.318). Stationary J per stage and BWT per stage are in Appendix E and Figure 9.

arm	hit at event g1/g2/g3/g4	FWT g1/g2/g3/g4	days to recover	gap to oracle at stage end g1/g2/g3/g4
continual (cgrpo, gated)	+.048/ − .050/ + .012/ − .008	−.020/ + .085/ + .130/ + .128	6/6/1/1	−.001/ − .010/ − .004/ − .014
continual (AWR+IPS, gated)	+.071/ − .019/ − .007/ − .004	+.012/ + .016/ + .014/ + .006	–/–/–/–	−.102/ − .107/ − .130/ − .155
frozen (days 1–5)	+.067/ − .041/ + .004/ − .024	+.060/ + .046/ + .057/ + .032	–/–/–/–	−.050/ − .075/ − .084/ − .116
static router	+.053/ − .033/ − .002/ − .033	+.034/ + .028/ + .032/ − .002	–/–/–/–	−.075/ − .093/ − .109/ − .150
router re-learned per stage	+.053/ − .041/ + .015/ − .026	+.034/ + .024/ + .066/ + .002	–/–/–/–	−.071/ − .076/ − .112/ − .103
`always_verify`	+.097/ − .037/ + .008/ − .021	+.075/ + .065/ + .080/ + .057	–/–/–/–	−.034/ − .056/ − .062/ − .090
`never_verify`	+.098/ − .043/ + .011/ − .006	+.115/ + .099/ + .117/ + .109	1/–/–/–	+.006/ − .022/ − .024/ − .038

Table 11: Whole-stream summary, default schedule (mild schedule in parentheses).

arm	mean BWT	mean FWT	regret, days 6–40
continual (cgrpo on replay, gated, learned λ)	+0.031 (+0.035)	+0.081 (+0.075)	1.262 (1.100)
continual (AWR+IPS, gated, $\lambda = 2$)	−0.006 (−0.005)	+0.012 (+0.002)	3.896 (3.668)
frozen (trained days 1–5)	0	+0.049 (+0.053)	2.597 (1.970)
SkillOrchestra-style static router	0	+0.023 (+0.025)	3.459 (2.902)
SkillOrchestra-style router, re-learned per stage	+0.019 (+0.017)	+0.032 (+0.014)	3.029 (3.057)
`always_verify`	0	+0.069 (+0.062)	1.988 (1.731)
`never_verify`	0	+0.110 (+0.100)	0.559 (0.394)

J, continual minus `never_verify`, by 8-day stage window: g0 −0.073, g1 −0.070, g2 −0.021, g3 +0.024, g4 +0.035; on the held-out sets −0.085, −0.087, −0.004, +0.016, +0.019. The served gap is positive from day 24 on and stays positive. Cumulative held-out regret to the oracle restricted to days 17–40 is 0.42 (continual) against 0.68 (`never_verify`); restricted to days 25–40 it is 0.22 against 0.50, and every other arm is further behind (frozen 1.60, re-learned router 1.74, static 2.07, AWR 2.14 from day 25). The learner's served J includes its $\epsilon = 0.1$ exploration mass, a tax the fixed policy does not pay, and it still wins after the crossover. The whole deficit in the whole-stream number is the 12 days on the uniform policy before the first promotion (finding 3) plus the repricing recovery (days 9–13, where the half-price mid tier makes `never_verify` the optimum before the learner has moved): `never_verify` is above the reference oracle at g0 and g1 (+0.023, +0.006 on the held-out sets at stage end) and falls to −0.022, −0.024, −0.038 at g2–g4 as capabilities it cannot use arrive. The post-crossover margin at g4 holds over 5 seeds (+0.031 ± 0.007, above); the stage-2+ capabilities are assumed parameters (Appendix D), so its size is the simulator's. The same analysis is finding 2 of the repository's `results/mas_continual/interpretation.md`.

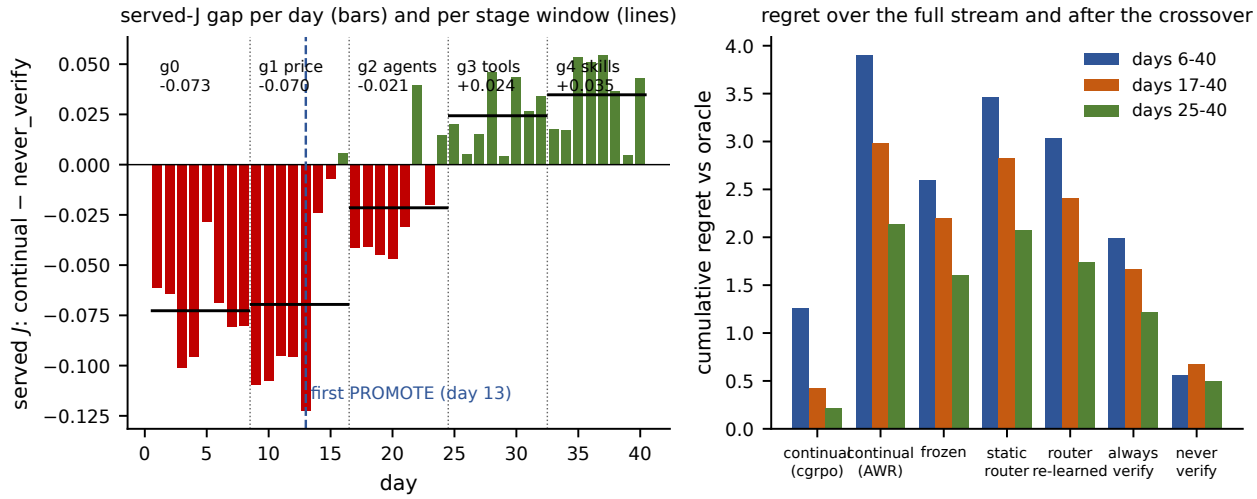


Figure 7: The crossover. Left: served J of the continual arm minus `never_verify`, per day (bars) and averaged per stage window (black lines), with the harness events (dotted) and the first promotion (dashed, day 13). Right: cumulative held-out regret to the oracle over the full stream (days 6–40) and restricted to after the agent-pool event (17–40) and after the tools event (25–40). Both panels from `days.jsonl`.

(3) The gate's price is larger than its benefit on this stream: 32 of 40 candidates reverted, 9 rightly and 23 as false alarms, first promotion on day 13. Figure 8 and Table 12. Every one of the 9 correct REVERTs was a budget violation (candidate true cost \$0.095–0.101 against $B + \text{tol} = \$0.095$) and no promoted candidate was bad. The DR cost estimate is the criterion that works: RMSE \$0.006 against the simulator's truth, and the set of candidates it stopped is exactly the set of over-budget candidates. The quality criterion produces the false alarms: 18 of the 32 REVERTs failed on quality alone with the candidate acceptable. At $n \approx 500$–1,000 logged episodes the DR quality 90% CI is 0.20 wide and covers the truth 63% of the time, so a candidate whose lower bound must reach the incumbent's point estimate minus 0.05 passes only when it is visibly better. During days 1–12 the served policy was therefore the uniform ϵ-policy while the learner was already good: the frozen arm, which is the learner's day-5 snapshot, scores 0.411 on g1 against the served incumbent's 0.247 on day 9. This is the same sample-size finding as Section 4.3: the cost side is certifiable at about 500 episodes a day, the quality side is not.

Table 12: Gate history and estimator accuracy against simulator truth (the candidate's true held-out value on the current stage), days after warm-up (35 gates, mean $n = 370$ episodes per gate).

	continual (cgrpo on replay)	continual (AWR+IPS)
candidates: PROMOTE / REVERT	8 / 32	14 / 26
REVERT right (candidate bad) / false alarm	9 / 23	0 / 26
PROMOTE right / missed (candidate bad)	8 / 0	14 / 0
mean per-decision ESS / n	0.54	0.25
mean DR quality CI width / cost CI width	0.200 / \$0.0079	0.653 / \$0.0144
DR quality: bias / RMSE / CI covers truth	−0.005 / 0.067 / 63%	+0.676 / 0.809 / 6%
DR cost: bias / RMSE / CI covers truth	+0.0003 / \$0.0061 / 54%	+0.0000 / \$0.0056 / 69%
per-decision SNIPS quality: bias / RMSE	−0.020 / 0.053	+0.178 / 0.187

(4) The AWR+IPS variant is the cautionary row. It was promoted 14 times and is the worst learned arm (stationary J 0.32–0.36) because the gate's sequential DR over-estimated its quality by +0.68 on average (coverage 6%). AWR's exponentiated inverse-propensity weights move the policy far from the behaviour in one full-batch step; the cumulative ratios (clipped at 50) then carry the estimate with a per-decision ESS of 25% and a bias the clip does not remove. The per-decision estimators' bias on the same candidates is +0.18, better but not a safe bar. The cgrpo candidates, trained on rollouts sampled from the ϵ-mixed served policy, stay close to the behaviour (ESS 54%) and the same estimator is nearly unbiased on them (−0.005). The gate is only as good as the overlap between candidate and behaviour, and a trainer that preserves that overlap is part of the safety story.

(5) Orchestrator-level BWT is bounded and mostly positive. Figure 9. The continual arm's BWT per stage is +0.111 (g0), +0.023 (g1), +0.009 (g2), −0.019 (g3); the mild schedule gives +0.120, +0.026, +0.002, −0.009. Later training did not undo earlier behaviour because the harness only grows: a capability that arrives at stage g has a zero feature slot and a zero logit before it, and the stage-g weights are still what runs where it is not legal. Contrast EvoHarnessBench's −34.7% BWT for agent-pool expansion under a fixed policy: here the pool expansion costs the fixed policies at the event (frozen hit −0.041, static −0.033 at g2) and they never recover (stationary J falls from 0.411 / 0.385 at g1 to 0.350 / 0.316 at g4), while the policy that moves loses nothing on the old stages.

(6) FWT is a property of the policy in service, not of anticipation. New capabilities arrive with zero logits, so the

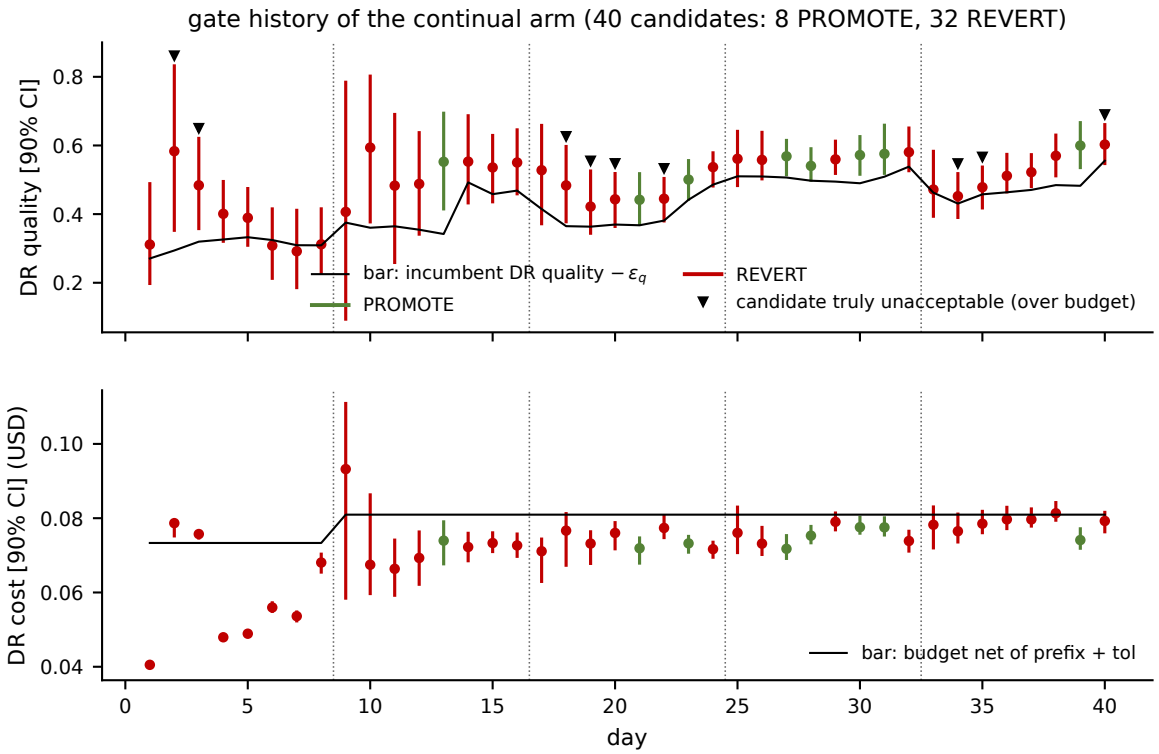


Figure 8: Gate history of the continual arm (from `gates.jsonl`). Top: DR quality with its 90% CI against the bar (incumbent's DR quality minus ϵ_q); bottom: DR cost against the budget net of the prefix plus tolerance. Green is PROMOTE, red REVERT; triangles mark candidates the simulator says were truly over budget. The cost criterion catches every bad candidate; the quality criterion rejects acceptable ones until they are visibly better.

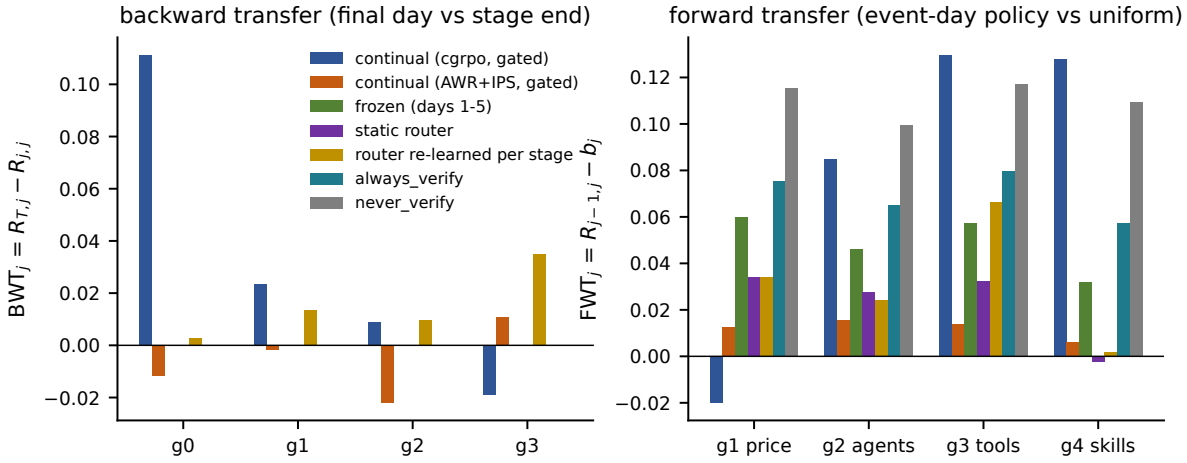


Figure 9: Orchestrator-level backward and forward transfer per stage, Eq. (12), computed from `days.jsonl`. The fixed arms have BWT 0 by construction. The continual arm's large g0 BWT is the day-8 incumbent being the uniform policy (finding 3), not transfer.

event-day policy is the previous stage's; the continual arm's FWT on g2–g4 (+0.085 / +0.130 / +0.128) exceeds the frozen arm's (+0.046 / +0.057 / +0.032) because it is a better policy, and the static router's FWT falls to −0.002 at g4 because its handbook prices unseen capabilities at a prior. Hits at the events are small for every arm except the price event, where the halved mid price lifts every mid-heavy policy by +0.05 to +0.10.

(7) The SkillOrchestra-style router is the weakest design-time baseline here, and the reason is diagnostic. The competence of an action is the mean outcome of episodes in which it was taken, which under per-step exploration is confounded by everything else in the episode: the handbook over-selects `verify@verify-1` (28% of decisions), an action the real run showed buys nothing. And λ_c is tuned by episode-level SN-IPS of a deterministic router against $\epsilon = 0.1$ traffic, whose ESS is 1–8 of 96, so the choice of λ_c is noise and the re-learned router's λ_c swings between 0 and 20 across stages. Re-learning per stage recovers some of the agent-pool loss (+0.043 at g4) at the price of three exploration days per stage, and still

ends 0.07–0.11 below the oracle. This is a statement about applying the competence-minus-priced-cost rule to multi-step orchestration traces with a per-step router, not about SkillOrchestra's own handbook learning from richer per-skill traces on its own benchmarks.

Mild-schedule sensitivity. Figure 10 and Table 11: the ordering is unchanged with smaller gaps (continual 1.10, `never_verify` 0.39, frozen 1.97, static 2.90). In the mild run a learner exceeds the oracle at g1 (gap +0.006), a reminder that the oracle is a reference, not a proven optimum.

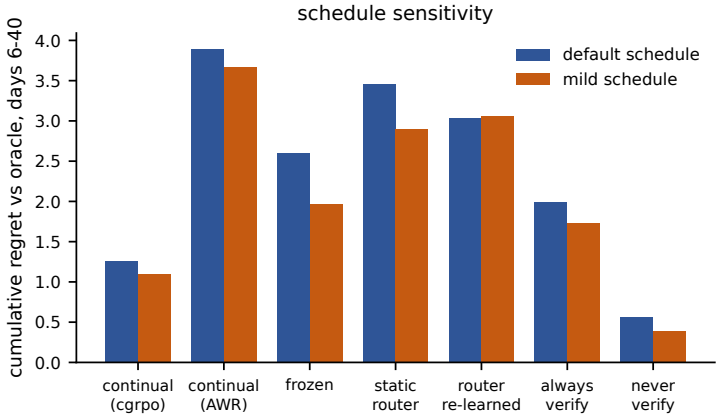


Figure 10: Cumulative regret under the default and the mild schedule. Same ordering, smaller gaps.

Verifier ablation: when does verification help?

Is "`never_verify` beats everything" an artefact of the measured verifier (uninformative: $P(\text{flag} \mid \text{wrong}) = 0.92$ against $P(\text{flag} \mid \text{correct}) = 0.96$) and of the 8-call cap that a verify-and-retry process fills? Four controlled variants of the same 40-day stream, 5 seeds each (seeds 0–4; arms: continual cgrpo, `always_verify`, `never_verify`, per-stage oracle): A, the committed configuration (verify-1, cap 8), which also supplies the seed error bars above; B, the informative verify-2 for every arm from day 1 (per-call $P(\text{fail} \mid \text{good rows})$ 0.58 → 0.114, recall unchanged); C, B plus cap 12; D, C plus verifier calls exempt from the cap. Served numbers are means over all 40 days (the fixed modes do not learn); J uses the stage oracle's $\bar{\lambda}_g$.

Table 13: Verifier ablation: `always_verify` minus `never_verify`, served, mean ± SE over 5 seeds.

variant	Δ expected quality	ΔJ	Δ cost / task	cap hits (always)	retries / ep. (always)	verifier calls / ep. (always)
A: verify-1, cap 8 (committed)	-0.033 ± 0.001	-0.036 ± 0.001	+\$0.0036	95%	0.89	2.94
B: verify-2 for every arm	-0.004 ± 0.001	-0.007 ± 0.001	+\$0.0030	91%	0.64	2.90
C: B + cap 12	$+0.029 \pm 0.001$	$+0.011 \pm 0.001$	+\$0.0148	9%	0.76	3.65
D: C + verifier exempt from cap	$+0.028 \pm 0.001$	$+0.010 \pm 0.001$	+\$0.0148	0%	0.77	3.64

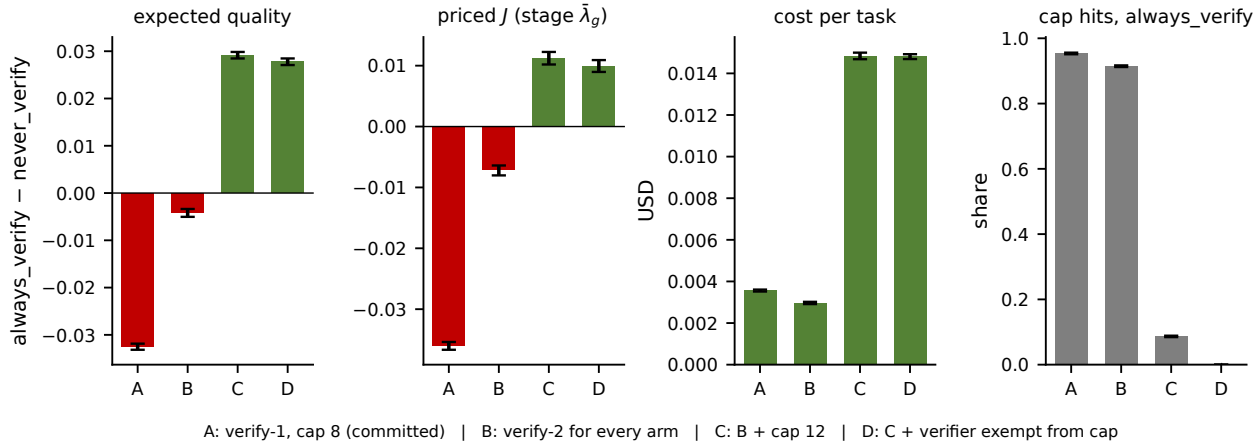


Figure 11: Verifier ablation (from the ablation summary file): `always_verify` minus `never_verify` in expected quality, priced J and cost, and `always_verify`'s cap-hit share, under the four variants; bars are means over 5 seeds with SE. The sign flips only when the verifier is informative *and* the retries do not come out of the plan's call budget.

Table 13 and Figure 11. Under the committed configuration verification *hurts*: `always_verify` is below `never_verify` by 0.033 ± 0.001 in expected quality and 0.036 ± 0.001 in J for +\$0.0036 per task, consistent with the real run's +1 task ($+0.019 \pm 0.067$ SE at $n = 50$). The informative verifier alone (B) closes the gap to -0.004 ± 0.001 but does not reverse it, because 91% of `always_verify` episodes still hit the cap and the plan is truncated. Raising the cap to 12 (C) flips the sign: $+0.029 \pm 0.001$ quality and $+0.011 \pm 0.001$ J for +\$0.0148 per task (+18%), with 3.65 verifier calls and 0.76 retries per episode and 9% cap hits (the stage-4 four-step plans). Exempting verifier calls from the cap (D) changes nothing beyond removing the last cap hits (+0.028 / +0.010): with one retry per sub-query a three-step plan already fits in 12 calls. Stated plainly: verification helps only when the verifier's fail is informative *and* the retry does not come out of the plan's call budget, and then it is worth about +0.03 accuracy for +18% cost, a priced gain of +0.01 J at the stage prices. Under C and D the fixed `always_verify` costs \$0.098–0.105 per task, above the learner's $B + \text{tol} = \$0.095$, so it can exceed the budget-constrained oracle at a stage (g1 in C: 0.443 vs 0.434); the learner's verify share rises from 0.03–0.09 (A) to 0.10–0.13 (B–D) of its decisions and no further, and within budget it still ends the last stage above `never_verify` under every variant (g4: A $+0.031 \pm 0.007$, B $+0.017 \pm 0.011$, C $+0.016 \pm 0.007$, D $+0.013 \pm 0.010$). The whole-stream regret ordering under C/D is `always_verify` 0.30 < `never_verify` 0.60 < continual 1.05–1.20: at a verifier worth using and a cap that does not truncate, the right fixed mode is `always_verify`, it is over budget, and the budgeted learner is third, its cold start still the whole deficit. Two caveats. The informative verifier's episode-level $P(\text{flag} \mid \text{correct})$ in the simulator is 0.71, not the 0.30 its per-call parameter was derived from, because correct episodes still contain empty and wrong steps that are rightly flagged; the per-call parameter is what the ablation sets. And the effect size rests on the assumed link from a step's latent quality to the answer.

Table 14: The real log (130 MAS episodes): what a flagged step's retry did. Observation kind before the flag (rows) against after the retry (columns).

before → after	rows	empty	error	invalid	respond	total
rows	26	2	5	0	0	33
empty	5	7	0	1	0	13
error	5	1	0	0	0	6
invalid	0	0	0	0	1	1

The real log gives the per-step number beside the per-episode one (Table 14): 107 verifier fails, 53 followed by a retry of the same sub-query (the rest hit the cap or the one-retry limit). The retry changed the observation kind in 10 of 53 cases (19%), always from empty/error to rows: a flagged empty, error or invalid step became rows in 10 of 20 cases, so the retry does repair malformed or empty steps; but 33 of the 53 retries were of a step that had returned rows, and 26 of those returned rows again. When the retried step was verified again it passed 11 of 47 times (23%) against 42 of 114 (37%) for first attempts. Paired with `never_verify` on the same tasks, the 44 `always_verify` episodes that retried flipped 6 wrong→right and 5 right→wrong, with 16 unchanged right and 17 unchanged wrong (25% flipped, symmetric). Per step and per episode say the same thing: the measured verifier's flag is not information about correctness.

What this does not show. Effect sizes are the simulator's: the stage-2+ capabilities are assumed, as is the link from a worker step's latent quality to the answer. One schedule plus a mild one; the all-arms run is one seed, with 5-seed error bars for the continual, `always_verify`, `never_verify` and oracle arms from the ablation of Section 4.7. The per-stage oracle is constrained GRPO on fresh exploration, not a proven optimum. Nothing here calls a model API; what a real-traffic run would require is in Section 5.

Platform and recipe controls on the failed run. We reproduced the phase-1b failure on three stacks. On TRL with DAPO's dynamic sampling and asymmetric clip, and on verl (Ray, vLLM) with its DAPO recipe, the policy learned the difficulty split that phase 1b never did (easy → mid, hard → frontier; quality 0.83 at cost 0.33 on the same 2,000 tasks), and the two stacks agreed within 0.001 in priced J: the platform contributes nothing measurable. NeMo-RL, run with the same recipe to 120 steps, lands on the same split and the same priced J (0.829 at cost 0.333, J 0.682; easy → mid, hard → frontier): the three stacks agree within 0.001. One systems detail surfaced in doing so: NeMo-RL's dynamic-sampling filter tests whether the group reward standard deviation is non-zero in floating point, so groups with exactly equal rewards often pass on rounding noise (58–62% of trained groups over the run had identical rewards, while its own mixed-group metric reports 100%); the split sharpened regardless, so the recipe's effect does not depend on a perfect filter. The control that settles the cause is TRL with phase-1b's own settings, no dynamic sampling, symmetric clip, and the price λ held fixed at 0.44 in the prompt and the reward: it learns a split too, sharp on easy tasks (mid 95%) and partial on hard ones (frontier 53%, mid 43%), at the same priced J (0.679; quality 0.796 at cost 0.268, inside the budget), with identical groups on 62% of steps on average after step 20. The phase-1b failure was therefore first the moving price, not the sampler and not the stack; dynamic sampling then sharpens the hard-task split (53% to 98% frontier) without changing priced J. With λ driven by dual ascent every step, the reward the policy optimised changed under it (the primal–dual orbit of Figure 4); held fixed, GRPO on default settings finds the mapping in 150 steps with no warm start. Dynamic sampling and the clip bought a crisper split, +0.002 in J, and a budget overshoot. All of these runs used one GPU, so the sharding strategies the stacks expose (FSDP, DTensor, Megatron) were nominal and no parallelism effect is measured. The open question for

a budgeted deployment becomes how to learn λ without destabilising the policy: slower dual steps, or alternating price and policy phases.

4.8 The same analysis at 300 tasks, and a second organisation

After the 50-task results above, the collection was resumed on the same seeded, category-stratified sample to 150 and then 300 tasks (900 episodes, every task on every tier), and a second collection of 150 tasks was run on the B2C organisation of the benchmark with the same protocol. Everything in Section 4.2 was re-run on each log with the same code and folds (`results/crmarena_300/`, `results/crmarena_b2c/`; the intermediate 127- and 150-task runs are kept and agree in every ordering).

	B2B, 300 tasks			B2C, 150 tasks		
held-out, 5-fold ($\lambda = 2$ per dollar)	quality	cost	J [fold SD]	quality	cost	J [fold SD]
fixed cheap	0.434	\$0.057	0.320 [0.063]	0.469	\$0.056	0.358 [0.078]
fixed mid	0.482	\$0.126	0.230 [0.026]	0.508	\$0.126	0.255 [0.109]
fixed frontier	0.614	\$0.131	0.353 [0.046]	0.638	\$0.143	0.352 [0.116]
AWR+IPS linear, greedy	0.596	\$0.080	0.437 [0.028]	0.598	\$0.081	0.435 [0.057]
per-category rule (train-fold argmax)	**0.644**	**\$0.079**	**0.486 [0.035]**	**0.636**	**\$0.081**	**0.474 [0.067]**
constrained, $B = \$0.10$, greedy of the mix†	0.644	\$0.103	0.521 [0.073]	0.677	\$0.105	0.549 [0.100]
per-task oracle	0.708	\$0.058	0.592 [0.032]	0.727	\$0.061	0.606 [0.068]

Table 15: CRMArena-Pro at 300 B2B and 150 B2C tasks: held-out tier policies. In both organisations the learned per-category policy beats every fixed arm held-out on quality and cost at once, at a margin of three fold SDs on B2B. $^{\dagger}J$ at each fold's learned $\bar{\lambda}$ (mean 1.22 per dollar on B2B, 1.35 on B2C); the B2C constrained policy sits \$0.0002 over the \$0.105 tolerance.

Four things are worth recording. (1) *The held-out ordering is stable across scale and organisation.* At 50 tasks no learned controller clearly beat the best fixed arm; at 150 the per-category rule led by a little over one fold SD; at 300 its J of 0.486 is 0.13 above the best fixed arm (frontier, 0.353) at three fold SDs, and the same rule leads on B2C, a different data schema and task mix, by 0.12 over its best fixed arm. The constrained controller reaches the frontier tier's quality at 22% less cost on B2B and exceeds it on B2C. (2) *The tier means keep moving as tasks are added* (B2B mid $0.533 \to 0.482$ from 150 to 300 tasks; frontier $0.616 \to 0.614$), which is why the held-out comparison, not the tier table, is the result. (3) *The gate's answer depends on the draw, and more data moves the rate, not the decision.* On the seed-0 realised one-arm log at 300 tasks the DR estimate for the all-task constrained policy is 0.607 [0.525, 0.668] in quality and \$0.095 [0.087, 0.113] in cost, and the gate REVERTs on the cost bound; the policy's exact cost is \$0.098 against a \$0.100 budget, so the candidate sits inside the budget by less than the tolerance and the interval cannot certify it. Over 200 redraws the promotion rate is 30% at 300 tasks (10% at 150, 6% at 50) and 8% on B2C, the DR cost interval has mean width 0.024 (0.030 on B2C) with 96% coverage of the exact value, and the DR quality estimate has RMSE 0.029. The width at 300 tasks matches the $1/\sqrt{n}$ projection made at 150 (0.023), and the promotion rate exceeded the projected ten to fifteen percent; the thousand-task estimate of Section 4.3 for certifying the \$0.005 margin stands. (4) *Repricing drift replicates on both organisations.* Held-out, a frontier price rise of $1.5\times$ costs the frozen policy -0.050 in priced value on B2B and -0.049 on B2C, of which a retrain recovers $+0.035$ and $+0.027$; halving the mid price is neutral ($+0.002$ on both) and the retrain gains $+0.031$ and $+0.030$ by moving traffic to mid; tripling the cheap price costs -0.038 and -0.047, and the retrained policy on B2B recovers $+0.044$ but leaves the budget. Every retrained candidate is REVERTed by the gate on the repriced realised log, on the quality bar recorded at the incumbent's promotion, the cost bound, or both: at this sample size the gate is the conservative element of the loop, which is its design.

4.9 Two further suites: τ-bench and EvoHarnessBench

The CRMArena-Pro collections are single-organisation, three- to four-turn tool-use tasks with a program-checked answer. To see whether the tier-level findings depend on that shape, the same counterfactual protocol (every task on every tier, propensity 1/3 at the first step, fixed continuation) was run on two suites with different shapes: τ-bench [44], a multi-turn conversation with a simulated customer under a policy document, and EvoHarnessBench [16], whose hosted service grades the final state of a live enterprise environment with hidden verifiers and grows the tool set across stages. Details are in Table 21.

τ-bench retail (60 tasks). The agent is the tier model; the user simulator is fixed to the cheap tier for every arm, its cost logged and excluded from the agent cost. Quality is the benchmark's own reward, pass^1 per task. Episodes are long, 13–15 agent turns, each resending the policy document and tool schemas, so the cost column is three to four times the CRMArena-Pro column and the tier spread is wider.

Two findings. First, the per-task headroom is the largest of any suite: the oracle reaches 0.867 at \$0.171, above the frontier tier's 0.717 at 35% of its cost, because the three tiers fail on different tasks (26 of 60 solved by all three, 52 by at least one). Second, none of it is reachable by a first-step controller, and the analysis says why: in τ-bench the task is revealed by the simulated user over the conversation, so at step 0 the agent has seen nothing but the domain, the feature

τ-bench retail, 60 tasks × 3 tiers	pass[1]	agent cost/task	held-out J ($\lambda = 2$) [fold SD]
fixed cheap	0.583	\$0.117	0.350 [0.135]
fixed mid	0.683	\$0.229	0.226 [0.154]
fixed frontier	0.717	\$0.487	−0.257 [0.176]
uniform behaviour	0.661	\$0.277	0.107 [0.146]
AWR+IPS linear, served	0.626	\$0.186	0.255 [0.135]
per-category rule / constrained ($B = \$0.28$), greedy	0.583 / 0.683	\$0.117 / \$0.229	= fixed cheap / = fixed mid
per-task oracle (max reward, cheapest among ties)	0.867	\$0.171	0.524 [0.080]

Table 16: τ-bench retail: fixed tiers, the oracle, and first-step policies (5-fold CV over task ids). 52 of 60 tasks are solved by at least one tier and 26 by all three. The first-step feature vector is the domain one-hot only, so every learned first-step policy collapses to a fixed arm.

vector is a one-hot, and the per-category rule and the constrained policy reduce exactly to the best fixed arm under their objective (cheap at $\lambda = 2$, mid under the budget). The realised-log gate on the all-task constrained policy REVERTs on the quality bar (DR 0.753 [0.613, 0.876] against a bar of 0.620) and 5% of redraws promote; with a binary reward and 60 tasks the DR quality interval is 0.30 wide. The decision that would capture the headroom is a mid-episode tier switch conditioned on the conversation so far, which the counterfactual protocol here does not log; it is the natural next collection on this suite.

τ-bench airline (49 tasks). The airline domain was collected on the same protocol to 49 of its 50 sampled tasks (the ledger cap bound before the last task).

τ-bench airline, 49 tasks × 3 tiers	pass[1]	agent cost/task	held-out J ($\lambda = 2$) [fold SD]
fixed cheap	0.449	\$0.099	0.251 [0.176]
fixed mid	0.551	\$0.262	0.027 [0.153]
fixed frontier	0.527	\$0.522	−0.516 [0.121]
uniform behaviour	0.509	\$0.294	−0.079 [0.121]
AWR+IPS linear, served	0.490	\$0.203	0.084 [0.134]
per-category rule / constrained ($B = \$0.29$), greedy	0.449 / 0.551	\$0.099 / \$0.262	= fixed cheap / = fixed mid
per-task oracle (max reward, cheapest among ties)	0.735	\$0.159	0.428 [0.064]

Table 17: τ-bench airline: the same analysis. 36 of 49 tasks are solved by at least one tier and 14 by all three; the mid tier beats the frontier tier at half its cost; the first-step policies again collapse to fixed arms, the realised-log gate REVERTs and 5% of redraws promote.

The airline domain is harder for every tier and repeats both findings: the oracle's headroom over the best tier is larger still (0.735 against 0.551, at 61% of the mid tier's cost), the mid tier beats the frontier tier outright, and no first-step policy can use any of it.

EvoHarnessBench, tools axis (50 tasks at each of stages 1, 2 and 4). The service exposes a harness that grows its tool set across six stages, $H_1 \subset H_2 \subset \ldots$; each task is run under the tool allowlist of its own stage, on all three tiers, with prompt caching on the fixed prefix (the cost column is therefore at cached prices and is not comparable to the other suites'). Score is the service's partial-credit grade, the fraction of hidden verifiers passed; pass requires all of them.

stage		cheap	mid	frontier	per-task oracle	oracle picks c/m/f
H_1 (50)	score	0.491	**0.702**	0.619	0.753	26 / 20 / 4
	pass	0.24	0.42	0.40	0.52	
	cost	\$0.062	\$0.146	\$0.298	\$0.136	
H_2 (50)	score	0.370	**0.699**	0.610	0.741	21 / 25 / 4
	pass	0.16	0.40	0.30	0.48	
	cost	\$0.044	\$0.114	\$0.227	\$0.106	
H_4 (50)	score	0.403	**0.643**	0.631	0.738	18 / 28 / 4
	pass	0.14	0.28	0.30	0.38	
	cost	\$0.056	\$0.147	\$0.262	\$0.138	

Table 18: EvoHarnessBench tools axis, 150 tasks with all three tiers (held-out and adaptation splits pooled; 34–36 held-out per stage). Score is the mean verifier pass rate; cost is per task at cached prices. The oracle picks the cheapest tier among those with the best score.

At every stage the mid tier beats the frontier tier on score at about half its cost, and the oracle beats every single tier at or below mid-tier cost, choosing the frontier tier for only 4 tasks in 50. The ordering is the same at each of the three harness sizes, so growing the tool set did not change which tier is worth buying. Two things this collection does not contain, stated so that they are not read into it: every task ran under its own stage's harness, so all held-out tasks

feed the benchmark's forward-transfer group and none of its backward-transfer cells (earlier-stage tasks re-run under later allowlists), and no controller was trained on these logs, since the stage-wise logs are of a size where Section 4.2 found first-step policies memorising. What the suite adds to the report is the tier-level replication under a growing harness, at list-independent cached prices, on a scorer that is not ours.

4.10 Three follow-up experiments: delegation credit, meta-learned initialisation, and a skill handbook

RAO-style delegation credit (negative). We added the node reward of Gandhi et al. [9] to the continual orchestrator's trainer: for a `delegate` decision, the per-decision reward is the episode's priced outcome plus λ_d times the mean success of the sub-call it spawned, with a leave-one-out baseline anchored on the root reward. Over five seeds on the committed simulator and schedule, delegation share rose at every stage (from 0.17/0.44/0.50/0.51/0.55 across the five stages without credit to 0.26/0.47/0.57/0.62/0.62 with λ_d=0.5) and served regret over days 6–40 rose from 0.94 ± 0.27 to 1.74 ± 0.18; stationary priced J fell by 0.02–0.04 at every stage after the baseline. Because the credit mode also changes the baseline, we ran the same mode with λ_d=0: regret 0.99 ± 0.24 and unchanged shares. The credit, not the baseline, buys over-delegation. The learner without credit already delegated to new specialists from the stage they appeared; a bonus outside the baseline turns that into a habit.

Meta-learned controller initialisation (positive, caveated). The learner's deficit against the fixed policy was its cold start. Following the meta-learned initialisation of Tandon et al. [36], we learned the controller's initial weights by a first-order meta-loop over the simulator's stage definitions g_0–g_3 (never g_4, never a held-out set; audited per run), then ran the continual arm from that initialisation. Over five seeds, served regret over days 6–40 fell from 1.06 ± 0.21 to 0.47 ± 0.08 (paired -0.59 ± 0.15) and over days 17–40 from 0.76 ± 0.18 to 0.43 ± 0.07; the learner now has lower regret than `never_verify` (0.74 ± 0.08) on the full stream. The gain comes from what is served before and between promotions, not from earlier promotion (first-promotion day unchanged at 5.2 in this arm set), and there is no gain on g_4, the one stage the meta-initialisation never saw. The caveat is structural: the initialisation was trained on the simulator's own stage definitions; on real traffic it would have to be learned from past harness generations only, and whether the gain survives that is untested.

Skill handbook for the cheap tier (null, with the control that showed it). The cheap tier scores 0.444 on the 50 real tasks because it fails schema reasoning, while the mid and frontier tiers solved many of the same tasks. We distilled per-category procedural notes from those successful SOQL episodes with the mid-tier model as writer (leave-one-task-out, so no task saw notes from its own solution; 49 writer calls) and served the cheap tier with the notes in its prompt. Quality rose to 0.581 ($+0.137 \pm 0.070$ paired). But the 14 tasks that received no notes improved by the same amount as the 36 that did, so we re-ran the cheap tier with no notes: 0.561, a difference from the skilled arm of $+0.020 \pm 0.067$. The handbook did nothing measurable; the original 0.444 was a low draw. What survives is the size of that draw: the same tier on the same 50 tasks moved by $+0.117 \pm 0.062$ between two runs a day apart, which is as large as the gap between tiers in Section 4.2, and a second, independent reason the gate of Section 4.3 refused to certify a margin at this sample size.

5 Discussion

The signal is the bottleneck. The same pattern appears in every experiment. On CRMArena-Pro the per-task oracle is far above every arm, but the only observable the controller has at step 0 is the category, and with 2–3 tasks per category it memorises (Section 4.2). The orchestration reward model finds that process features carry no out-of-fold signal while the category alone has AUROC 0.82 (Section 4.5); the verifier flags nearly everything; the self-refinement probe barely sees whether the cheap answer is wrong (AUROC 0.65), so the best priced policy is "mostly do not refine" (Section 4.6). The LLM controller's GRPO stage had no advantage signal because its rollout groups were identical, and what it did learn was the price (Section 4.4). In the continual study the gate's quality criterion has too little signal at $n \approx 500$ to tell an acceptable candidate from the incumbent, while the cost criterion, whose signal is a sum of logged dollars, works (Section 4.7). The learners are not the limiting factor; the observables and the sample sizes are. On this benchmark the right orchestration signal is known before the first call, and that argues for investing in observables (a verifier trained on logged (task, answer, score) triples; the mid arm's own answer as a check) before investing in policy classes.

When does a learned orchestrator earn its keep? Only when a harness change moves the right fixed policy. On the stream of Section 4.7, the capability-adding stages are where the continual arm's stationary J exceeds the best fixed mode, and they are also where the fixed modes' gap to the oracle grows ($-0.006 \to -0.038$ for `never_verify`; frozen $0.411 \to 0.350$). A repricing moves the fixed policy's cost but not its ranking; a new specialist or tool moves the ranking, and only a policy that can see the new capability can use it. If the harness is stable and someone knows the right fixed mode, the learner's whole-stream regret (1.26) is its cold start, its recovery days and its gate delays, and the fixed mode (0.56) wins; once the pool has grown past what the fixed mode can use (from day 25: 0.22 against 0.50; g4 margin $+0.031 \pm 0.007$ over 5 seeds),

the learner wins while paying an exploration tax the fixed mode does not. And which fixed mode is right depends on the harness: with an informative verifier and a cap the retries do not consume, it is `always_verify`, over budget, and the budgeted learner is third (Section 4.7). The decision to deploy a learned orchestrator is therefore a decision about how often the harness will change in ways that re-rank the fixed policies, which is exactly what the design-time harness-evolution work predicts will be often.

Gate power against day size. With binary outcomes and sequential ratios, the DR quality interval is about ± 0.04 or wider at $n \approx 500$ ($\epsilon_q = 0.05$ was needed because the core default of 0.03 rejects the incumbent against itself at this n), while the cost interval is \$0.008 wide and its criterion is the binding one. Two things follow for a real deployment: certify cost and treat quality as a floor against regression, not as an improvement test; and keep the candidate close to the behaviour (the cgrpo arm's ESS of 54% against AWR's 25%), because the estimator's bias, not its variance, is what promoted the bad arm.

What Phase B would need. The same loop on the real orchestrator: days of 20 b2b tasks, the learner trained with AWR+IPS on the day's log (no replay exists on the real harness) and gated with the per-decision estimators, held-out evaluation by re-serving a fixed 20-task set per stage. Stages that can be induced for real: a repricing (free, costs recomputed from logged tokens); a better verifier (swap its prompt or tier); a new worker or answer backbone (any priced model); a worker swap by tier; a schema-lookup tool (one describe call) or a local aggregation tool; skills as steering text (tokens only). Such a run needs on the order of 20 tasks a day for 20–40 days across two to four induced stages, with a fixed 20-task held-out set re-served at each stage: about 460–900 episodes for one served policy, and about 2,500 if the two fixed baselines are served on the same days. A \$0.005 margin on a \$0.09 budget needs about 1,000 logged episodes to certify, so at 20 tasks a day the cost criterion would run on a rolling window and still be underpowered, and the quality criterion at $\epsilon_q = 0.05$ needs on the order of 500 episodes. A run of that size measures the mechanics (what the learner serves, what the gate does, the frozen policy's hit at each stage) rather than certifying improvements.

6 Limitations and threats to validity

- **Fifty tasks.** The real-harness evidence is one b2b collection of 50 tasks (2–3 per category); every held-out number is a mean over 10 tasks per fold, one task is 0.1 of quality, and the across-fold SDs are as large as the differences between policies. The MAS run has 130 episodes on the same 50 tasks; the SSR run 41 tasks. The negative results for the linear controller, the ORM and the refinement gate are statements about this sample size and these observables, not about the approaches.
- **Seeds.** The simulator pipeline reports 5 seeds; the CRMArena folds one split, and each GPU run ran once; the continual study's all-arms run is one seed per schedule, and only the verifier ablation (continual, the two fixed modes and the oracle; Section 4.7) reports 5-seed error bars. The router and AWR arms carry no seed variance.
- **Simulators calibrated to the project's own logs.** The sim of Section 4.1 is hand-written. The continual simulator's process statistics are measured on 130 episodes, but the link from a step's latent quality to the answer and every capability that arrives after stage 0 are assumed (Appendix D); the transfer effect sizes are the simulator's, not CRMArena's. The calibration check reproduces the fixed modes' cost and call counts but lands accuracy at 0.446/0.459 against 0.480/0.460, because the simulated verifier is uninformative, as the measured one is.
- **Assumed stage parameters.** $P(\text{good} \mid \text{rows}) = 0.60/0.75/0.85$ by tier; coverage-to-answer logit slope 1.5; task-level logit noise sd 0.4; `verify-2` at $P(\text{flag} \mid \text{correct}) = 0.30$ with the same recall; `data-4` at rows 0.85 / good 0.80; `answer-4` mid-grade on 8 categories; `schema_lookup` removes error mass from the next step; `aggregate_calc` adds +0.34 coverage on aggregation categories; the three skills add +0.15 $P(\text{good} \mid \text{rows})$ on their categories or re-plan into 3–4 sub-queries; planner v2 plan length $(0.0, 0.1, 0.4, 0.5)$. Tools and skills are priced as a verifier-sized cheap call (711/62 tokens).
- **The SkillOrchestra baseline is the literal rule, not their system.** Competence minus λ_c times mean logged cost, over task categories as skills, learned from per-step exploration traces of a multi-step orchestrator. Their handbook learning from richer per-skill traces on their own benchmarks is not reproduced.
- **The oracle is a reference.** Constrained GRPO on fresh exploration with more iterations, not a proven optimum; regret can be slightly negative.
- **Gate design.** The sim gate fits its direct model in-sample; the bandit gate cross-fits by task; the continual gate uses the sequential DR with cumulative ratios clipped at 50, whose bias on far-from-behaviour candidates is what finding (3) exposes.
- **Data terms.** CRMArena-Pro is CC BY-NC 4.0 (research use), fetched at setup and not copied; the committed trace is a derived research artefact under the same terms (Appendix G).

7 Related work

Cost-aware routing. FrugalGPT [5] cascades models by a learned score to cut cost; RouteLLM [26] learns a router between a strong and a weak model from preference data; xRouter [29] trains a cost-aware orchestration system with

reinforcement learning. These choose a model per query at the first turn; this report prices every decision point inside a frozen harness, learns the price as a dual variable against a budget, and gates deployment off-policy.

Self-evolving agent systems and the deployment seam. A recent position paper from the AReaL group [42] argues that enterprise self-evolving agents are blocked not by RL algorithms but by systems: a step-level episode-log protocol, a data proxy that turns production workloads into governed learning material, and a unified *evolution control plane* that selects, from episode-log statistics, drift indicators and cost, among memory updates, skill updates, harness edits, weight updates, rollback or no-op, each behind replay-first evaluation, regression tests and canary rollout. That is the architecture this report instantiates in part: the gate of Section 4.3 is a measured, interval-valued form of their replay-first check; priced backward and forward transfer are the drift indicators they call for; and the continual orchestrator of Section 4.7 is a per-request learner inside the harness surface they name, with the harness-edit surface left to the slow loop. Their prototype realises the weight-update path by redirecting deployed agents' inference calls through managed workers, the same seam that Agent Lightning [24] uses to make training harness-agnostic through a proxy that emulates the inference APIs; placing the controller in that proxy is the deployment path we would take, since tier choice, verification and stopping are decisions at the API boundary, while replan and delegation still need a hook inside the orchestrator. Production precedent for the cadence exists: online RL after deployment has shipped with checkpoints refreshed from user interaction every few hours.

Recursive delegation and priced effort. Recursive Agent Optimization [9] trains agents that spawn instances of themselves through a `launch_subagent` call, with a local node reward equal to the node's own success plus λ_d times the *mean* success of its children (averaging rather than counting, which removes the incentive to spawn), a leave-one-out baseline anchored on the root reward, and depth-level inverse-frequency weighting so that deep sub-tasks do not dominate the update. Four points carry over. First, that node reward is the principled process reward for the delegate action of Section 4.5, where our process rewards were definitional; we adopt its form in the continual study. Second, depth weighting is the recursive counterpart of the per-decision estimators used here. Third, delegation in RAO is unpriced: depth and step caps stand in for a budget, so adding $\lambda \cdot \text{cost}$ and $\mu \cdot \text{latency}$ to the node reward (their sequential case runs $18\times$ slower in wall-clock) is the one-line extension that turns the caps into a learned price, and the comparison it sets up is in-weights delegation against controller-side delegation at equal cost. Fourth, RAO's generalization from medium to hard tasks concerns the learned decomposition skill, not drift of the harness around it, which is what the transfer metrics here measure. On the shaping of the price itself, the Composer 2 report describes a nonlinear effort penalty over tokens, tool calls and turns whose slope is steeper on easy tasks than on long-horizon ones; the objective here uses a single linear λ, and a difficulty-conditioned price, λ per task category or a concave cost curve, is a direct extension that the 50-task log is too small to fit.

Contextual bandits in production. The Decision Service [1] and Horizon [10] established the logging discipline (propensities with exploration), the offline evaluation [4, 22] and the promotion workflow that the gate here follows; the estimators are those of Dudík et al. [8], Jiang and Li [14], Swaminathan and Joachims [35], Thomas and Brunskill [38], with cross-fitting after Chernozhukov et al. [7].

Test-time compute and self-refinement. Socratic Self-Refine [32] decomposes an answer into verifiable sub-questions and rewrites where confidence is low; the uncertainty signals (verbalised confidence, sample agreement) follow Bakman et al. [3]. Section 4.6 treats how much of it to buy per request as a priced decision and finds that, on a CRM benchmark, a better tier up front dominates.

Process verification and reward modelling for multi-agent systems. MAS-ProVe [40] studies process-level verification of multi-agent executions; reward modelling for orchestration [39] scores an orchestration plan before any sub-agent runs, with a Bradley–Terry model trained on preference pairs mined from execution logs, so that best-of-N selection and orchestrator training avoid full rollouts; MAS-Orchestra [15] trains a holistic orchestrator with an outcome-only reward and asks when a multi-agent system beats a compute-matched single agent; ToolOrchestra [34] trains a stepwise orchestrator over tools and models with a composite reward that includes an efficiency term, the closest prior use of a priced orchestration reward, though unpriced at the per-decision level and without an off-policy gate. Section 4.5 measures all three on one log and finds the verifier uninformative, the process features without out-of-fold signal, and the single agent ahead.

Harness evolution and continual-learning metrics. The design-time line [2, 6, 12, 21, 45, 46] and EvoHarnessBench [16] evolve or grow the harness; SkillOrchestra [41] learns a static competence router. The BWT/FWT definitions are those of Lopez-Paz and Ranzato [23], applied here to a priced objective at the orchestrator level.

Offline and constrained RL. CQL [19] and IQL [18] are the standard pessimistic offline learners; AWR [27] is the simpler weighted-regression trainer used here with inverse propensities; GRPO [31] supplies the group-relative advantage; the learned price follows RCPO [37] and the PID-Lagrangian [33], with iterate averaging after Nedić and Ozdaglar [25].

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A Experimental details

One table per experiment. Values are taken from the run metadata files, the command-line defaults of the scripts that produced each result, and `results/llm/backends/COMPARISON.md`; a setting that none of these records is marked "not recorded". Costs in the CRMArena-Pro rows are token counts times the list prices given in Table 20; simulator costs are in simulator units.

Table 19: Simulator pipeline (Section 4.1; `scripts/run_v0.sh`, `scripts/run_sim_seeds.sh`).

setting	value
harness	simulator v_1; drift event v_2 plus reprice(mid, 1.2); at most 4 steps; 12 actions
policy class	softmax-linear, Eq. (4); 48 features; logit clip $L = 3$; served exploration $\epsilon = 0.05$
trainer	constrained GRPO on replay (`pace.train_offline --method cgrpo`; NumPy, no RL library)
group size / batch	$G = 8$ continuations per state; 32 logged states per iteration
iterations	300 on v_1; 300 for the v_2 retrain, warm-started from the v_1 policy
learning rate	1.5 (code default)
ratio clip; KL	none; none (on-policy group-relative gradient, Eq. (5))
reward	realised 0/1 outcome minus λC
λ learning	dual step $\eta = 0.1$, cost EMA $\rho = 0.9$, damping $k_p = 2.0$ (Eq. (3); inferred from the logged λ and cost EMA of the first iterations, not stored as settings); served $\bar{\lambda}$ and policy averaged over 30 snapshots from the second half
budget	$B = 0.30$
behaviour data	v_1: 3,000 episodes, uniform behaviour; v_2: 6,000 episodes served by the frozen v_1 policy with $\epsilon = 0.3$; propensities logged with exploration
gate	IPS, SNIPS, DR (in-sample bucketed direct model); cumulative ratio clip 50; 200 bootstrap resamples, 90% percentile intervals; $\epsilon_q = 0.03$, tol 0.02; v_2 bar from the v_1 promotion
ground truth	Monte Carlo over 20,000 fresh tasks, expected success probability
seeds	replicate $s \in \{0, \ldots, 4\}$: collect / train / collect / train seeds $10s{+}1, \ldots, 10s{+}4$; gate bootstrap and ground-truth seed s ($s = 0$ is `run_v0.sh`)
hardware	CPU

Table 20: CRMArena-Pro collections and the offline analyses on them (Sections 4.2, 4.3, 4.5, 4.6, 4.8).

setting	value
benchmark	CRMArena-Pro, b2b and b2c orgs, evaluator in `aided` mode (exact match, F1 for fuzzy categories, privacy judge)
tiers and list prices	cheap `claude-haiku-4-5`, mid `claude-sonnet-5-5`, frontier `claude-opus-5-5`; USD per million input / output tokens: 1.00 / 5.00, 2.00 / 10.00, 4.00 / 20.00; no prompt caching
counterfactual collection	b2b: 50 tasks (2–3 per category, all 22), extended on the same seeded, category-stratified sample to 150 and then 300; b2c: 150 tasks on the same protocol; every task on all three tiers; fixed continuation on the chosen tier; seed 0
propensity	1/3 at step 0 (uniform over tiers), 1 after
episode limits	12 agent turns (`ANSWER` forced at the last); 2,048 output tokens per agent turn, 1,024 for the forced answer; observations truncated to 20,000 characters; reasoning effort `low` for mid and frontier; no sampling parameters sent
offline learning on the log	AWR+IPS at $\lambda \in \{2, 5, 10\}$ per USD; constrained GRPO on the lookup table at $B = \$0.10$ with learned λ; 300 AWR epochs and 300 constrained-GRPO iterations (code default); 5-fold category-stratified CV over task ids, seed 0
gate on the realised log	bandit DR with cross-fitted direct model; 200 bootstrap resamples; 200 redraws of the one-arm log; $\epsilon_q = 0.03$, tol \$0.005
multi-agent run	same 50 tasks; planner cheap (600 output tokens, 1–3 sub-queries), data worker mid (one agent turn per sub-query), verifier cheap (200 output tokens, output shown truncated to 3,000 characters), answer mid; at most 8 role calls, 1 re-plan, 1 retry per sub-query; modes `always_verify` (50 episodes), `never_verify` (50), uniform exploration with logged propensities (30)
orchestration reward model	logistic regression and a 16-unit MLP; out-of-fold over tasks
self-refinement run	initial answers: the cheap-tier episodes of the counterfactual log; decompose on cheap (600 output tokens, at most 3 claims); $k = 2$ re-solve samples per claim, at most 3 turns each; refine (512 output tokens) on cheap and on mid; 42 tasks probed, 41 complete; controller: softmax-linear over 36 features, exact policy gradient on the logged tree, $B = \$0.10$ or $\lambda \in \{2, 5\}$, 5-fold CV over tasks
hardware	CPU client; models served by the provider's API

Table 21: The two further suites (Section 4.9; `pace.experiments.taubench`, `pace.experiments.evoharness`).

setting	value
τ-bench	retail domain, 60 tasks, and airline domain, 49 of 50 sampled tasks (the ledger cap bound before the last); seeded shuffle of the task ids (seed 0); agent on the tier model, user simulator fixed to the cheap tier for every arm with its cost logged separately; reward the benchmark's own pass[1] from its environment checker; at most 30 agent turns, 2,048 output tokens per turn, 512 for the user turn; reasoning effort `low`; same tiers and list prices as Table 20, no prompt caching
τ-bench offline analysis	as Table 20: AWR+IPS at $\lambda \in \{2, 5, 10\}$, constrained GRPO on the lookup table at $B =$ the uniform behaviour's cost (\$0.277), 5-fold CV over task ids; first-step features are the domain one-hot
EvoHarnessBench	hosted service, tools axis, stages 1, 2 and 4, 50 tasks each from a seeded round-robin over (domain, split, cohort) groups covering both the held-out and the adaptation split; each task under its own stage's tool allowlist on all three tiers; score the service's hidden-verifier pass rate, pass = all verifiers; at most 20 agent turns, 2,048 output tokens; reasoning effort `low`
EvoHarnessBench pricing	prompt caching on the system prompt (domain policy, action protocol, stage tool schemas) and the last message; cache writes priced at 1.25× and reads at 0.1× the input price; the cost column is therefore at cached prices and is reported separately from the other suites
hardware	CPU client; models served by the provider's API

Table 22: LLM controller, GRPO with LoRA (Section 4.4; `results/llm/run_meta_phase1*.json`). The CRMArena-Pro run `crm_phase2` uses the `phase1c` recipe in lookup mode on fold 0.

setting	`phase1`	`phase1b`	`phase1c`
model	Qwen3-1.7B, bf16, LoRA $r = 16$, $\alpha = 32$, dropout 0.05 on q, k, v, o, gate, up, down projections		
library	TRL 1.15.0 (loss type `dapo`, group-scaled rewards), Transformers 5.19.0, PEFT 0.21.2, PyTorch 2.14.1		
prompts × group size	8×8	8×8	4×16
completions per step; micro-batch	64; 16	64; 16	64; 16
steps per generation	4	4	4
max completion length	16 tokens	16 tokens	16 tokens
learning rate	10^{-5}	10^{-5}	10^{-5} (SFT 2×10^{-4})
clip (low / high)	0.2 / 0.2 (library default)	0.2 / 0.2 (library default)	0.2 / 0.2 (library default)
KL coefficient; reference	0.02; base model	0.05; base model	0.05; SFT model (adapter merged)
entropy coefficient	0.0 (code default)	0.01	0.01
temperature	1.0 (code default)	1.0	1.2, raised to 1.5 at step 20
GRPO steps	300	300	200, after 150 SFT steps
SFT data	–	–	oracle labels on 2,000 tasks at shown prices $\{0.0, 0.1, \ldots, 2.0\}$; 1,000 held-out prompts
λ: dual step / EMA / k_p	0.1 / 0.9 / 2.0	0.02 / 0.95 / 0.5	0.02 / 0.95 / 0.5, $\lambda_0 = 0.5$, shown in prompt
budget	$B = 0.30$	$B = 0.30$	$B = 0.30$
training tasks	2,000 seeded simulator tasks (seed 0), legal-action order shuffled per task		
evaluation set	2,000 fresh simulator tasks, seed 777, greedy decoding		
seeds; hardware	one run each, seed 0; one NVIDIA A100-SXM4-40GB		

Table 23: Platform and recipe controls on the phase-1b setup (Section 4.7, last paragraph; `results/llm/backends/COMPARISON.md`). Shared by all arms: Qwen3-1.7B with LoRA $r = 16$, $\alpha = 32$; temperature 1.0; KL coefficient 0.05 against the base model (k3 estimator in the loss for verl and NeMo-RL); entropy coefficient 0.01; learning rate 10^{-5}; at most 16 completion tokens; price fixed at $\lambda = 0.44$ in the prompt and the reward; the phase-1c task pool (2,000 tasks, seed 0); one A100-40GB; one run each. Evaluation for every adapter: one evaluator (HF `generate`), greedy and sampled at $T = 1$, on the 2,000 seed-777 tasks with the $\lambda = 0.44$ prompt.

setting	TRL, fixed price	TRL + dynamic sampling	verl, DAPO recipe	NeMo-RL, DAPO recipe
version	TRL 1.15 (phase-1b stack)	TRL 1.15, HF `generate`	verl 0.9.1, vLLM 0.24	NeMo-RL NGC v0.7.0, vLLM 0.20
prompts × group size	4×16 (code default)	4×16	4×16	4×16
clip (low / high)	symmetric; 0.2 / 0.2 (code default)	0.2 / 0.28	0.2 / 0.28	0.2 / 0.28
dynamic sampling	no	yes (up to 8 generation rounds)	yes (`filter_groups`)	yes
loss aggregation	token mean over the accumulated batch, `dapo` (library default)	token mean over the accumulated batch, `dapo` (library default)	token mean	token level
steps completed	150	186 (adapter at step 150 evaluated)	117 (checkpoint at step 100 evaluated)	25 (training curves only, not evaluated)

Table 24: Continual orchestrator (Section 4.7; `pace.experiments.mas_continual`). Simulator parameters are in Appendix D.

setting	value
stream	40 days × 96 seeded tasks per day, the same tasks for every arm; warm-up 5 days
stage schedule	g1 reprice (day 9), g2 agents (day 17), g3 tools (day 25), g4 skills (day 33); default and mild schedules (Table 9)
held-out set	150 fixed tasks per stage, simulator expected quality, never seen by the gate
policy class	softmax-linear over 18 actions and 88 features; served $\epsilon = 0.1$; zero feature slots and logits for capabilities not yet present
trainer (continual)	constrained GRPO on replay from the last 5 days of logged states; 60 iterations per day, batch 16, group 6; warm start from itself; iterate averaging
trainer (AWR variant)	AWR+IPS, fixed $\lambda = 2$ per USD, 200 epochs
learning rate; ratio clip; KL	1.5 constrained GRPO, 2.0 AWR (code default); none; none
λ learning	learned against $B =$ \$0.090 per task; dual step 20, cost EMA 0.9, damping $k_p = 40$, per USD of cost gap (code default)
gate	sequential DR, cumulative ratios clipped at 50; 200 bootstrap resamples; $\epsilon_q = 0.05$, tol \$0.005; log since the last harness event, at most 5 days; budget net of the policy-independent prefix
per-stage oracle	constrained GRPO on 200 fresh exploration episodes, 300 iterations, batch 24, group 8, no gate
SkillOrchestra-style router	handbook from warm-up exploration (days 1–4), λ_c chosen on day 5 by episode-level SN-IPS; re-learned variant: 3 days at $\epsilon = 0.3$ per stage, then refit
prices	list prices of Table 20, scaled by the stage's repricing
metrics	recovery tolerance 0.02; stationary window 4 days; regret over days 6–40
seeds; hardware	all arms seed 0 (default and mild); verifier ablation and the follow-ups seeds 0–4; CPU

Table 25: Simulator pipeline, per seed (`results/sim_seeds/RESULTS.md`). DR quality with its 90% CI; "cost hi" is the DR cost upper bound against $B + \text{tol} = 0.32$; truth is Monte Carlo over 20,000 tasks (for v_2, the adapted policy); recovery as served after the gate and for the adapted policy.

seed	$\bar{\lambda}$	gate v_1 DR quality	cost hi	decision	truth v_1 q / c	gate v_2 DR quality	cost hi	decision	truth adapted q / c	priced-BWT	recovery (served)	recovery (adapted)
0	0.343	0.851 [0.815, 0.887]	0.309	PROMOTE	0.880 / 0.303	0.873 [0.854, 0.889]	0.311	PROMOTE	0.871 / 0.312	−0.045	+0.033	+0.033
1	0.379	0.880 [0.848, 0.911]	0.310	PROMOTE	0.877 / 0.310	0.871 [0.851, 0.892]	0.292	PROMOTE	0.864 / 0.292	−0.049	+0.042	+0.042
2	0.278	0.901 [0.878, 0.921]	0.313	PROMOTE	0.877 / 0.309	0.879 [0.860, 0.895]	0.304	REVERT	0.869 / 0.304	−0.041	−0.000	+0.034
3	0.443	0.883 [0.845, 0.905]	0.313	PROMOTE	0.878 / 0.306	0.855 [0.825, 0.881]	0.290	REVERT	0.864 / 0.288	−0.057	+0.001	+0.051
4	0.346	0.903 [0.870, 0.925]	0.308	PROMOTE	0.877 / 0.302	0.867 [0.847, 0.887]	0.296	REVERT	0.866 / 0.296	−0.042	+0.000	+0.033

B Prompts

The templates below are copied from the code. Text in braces, `{like_this}`, is filled at run time; everything else is literal. Where a prompt is assembled by a function rather than stored as a constant, the listing is the exact output of that function on a seeded example, with the variable fields named in the paragraph heading. The CRMArena-Pro agent's own system prompt (the benchmark's ReAct prompt with the rendered org schema and metadata, built by `system_prompt()` in `pace/harness/crmarena.py` from the vendored `crm_sandbox.agents.prompts`) is the benchmark's text under its licence and is not reproduced; the planner and the worker run inside that system prompt, and the verifier and the self-refinement roles do not.

Controller, system message (`pace/llm/env_text.py`, `SYSTEM`).

```
You are the routing controller of an agent harness. You pick which model tier runs the next operation. Reply with exactly one action tag and
    nothing else.
```

Controller, simulator user message (`sim_prompt`; rendered for sim task 0, seed 0, $B = 0.30$; difficulty proxy, state, prices, the shuffled legal list and the example action vary per task).

```
Task: answer one question. Observed difficulty: hard (a noisy estimate, right about 85% of the time).
State: step 0, evidence 0, verified no, cost so far 0.000.
Budget: average cost per task must stay at or below 0.30.
Price of one ANSWER call by tier: cheap 0.020, mid 0.100, frontier 0.400. RETRIEVE and VERIFY cost 0.5x (RETRIEVE adds a 0.005 search fee),
    ESCALATE costs 1.5x.
Ops: ANSWER = answer now; RETRIEVE = fetch evidence; VERIFY = check with a judge; ESCALATE = answer now with extra compute.
After a non-final op the harness answers on the same tier.
Legal actions: mid ANSWER, cheap ESCALATE, frontier VERIFY, frontier RETRIEVE, mid ESCALATE, frontier ANSWER, cheap RETRIEVE, mid VERIFY, mid
    RETRIEVE, cheap ANSWER, frontier ESCALATE, cheap VERIFY.
Choose the action that maximizes expected answer quality within the budget.
Reply with exactly one tag, for example <action>frontier VERIFY</action>.
```

Controller, price-conditioned variant (phase 1c and the backend runs): the lines that differ, at $\lambda = 0.44$ (shown rounded to 0.1).

```
Price of cost: λ=0.4 (your reward is quality − λ·cost).
Choose the action that maximizes expected quality − λ·cost.
```

Controller, CRMArena-Pro lookup mode (`lookup_prompt`; $B = \$0.10$, typical episode costs from the 50-task log, legal list unshuffled; an optional `Query:` line follows the first line when the task text is included).

```
Task: a CRM agent task (org {org}, category {category}, scored by {reward_metric}).
Budget: average cost per task must stay at or below $0.100.
Typical cost of a full episode by tier: cheap $0.0680, mid $0.1196, frontier $0.1642.
CONTINUE = run the agent on that tier until it answers.
Legal actions: cheap CONTINUE, mid CONTINUE, frontier CONTINUE.
Choose the tier that maximizes expected answer quality within the budget.
Reply with exactly one tag of the form <action>TIER OP</action> naming one legal action.
```

Planner, user message (`pace/experiments/mas/roles.py`; system message: the benchmark's ReAct prompt).

```
Request: {query}

[MAS PLANNER] You are the planner of a CRM data team. Decompose the user's request into 1 to 3 sub-queries that a data worker will run one at a
      time against the Salesforce instance described above. Each sub-query has a `goal` (what it should establish) and a candidate `soql` (SOQL
      or SOSL text; leave it empty if the worker should write it). Order them so later sub-queries can use earlier results. Reply with JSON only:

{"subqueries": [{"goal": "...", "soql": "..."}], "rationale": "one sentence"}
```

Planner, re-plan user message.

```
Request: {query}

Evidence so far:
- {evidence item, at most 400 characters}
...

[MAS PLANNER] You are re-planning. The evidence gathered so far is listed below, with the verifier's findings. Produce the REMAINING sub-queries
       (1 to 3) needed to answer the request; do not repeat sub-queries whose results were verified. Same JSON format.
```

Worker, steering text appended to the agent's running ReAct context (the `Candidate SOQL` line only when the plan gave one).

```
[MAS WORKER] Sub-query {k} of {n}: {goal}
Candidate SOQL: {soql}
Run exactly one query now with <execute>...</execute> (adapt the candidate if needed).
```

Worker, steering text on a retry after a verifier fail.

```
[MAS WORKER] Sub-query {k} of {n}: {goal}
Candidate SOQL: {soql}
The verifier found a problem with your previous attempt: {reason}. Fix the query and run it again.
Run exactly one query now with <execute>...</execute> (adapt the candidate if needed).
```

Verifier, single user message (no system message).

```
Request: {query}
Sub-query goal: {goal}
Query run: {executed query}
Output: {Salesforce output, truncated to 3,000 characters}

[MAS VERIFIER] You verify one step of a CRM data worker. Given the user's request, the sub-query goal, the query that was run and the Salesforce
       output, decide whether this step's result is usable: well-formed, non-empty when the goal needs rows, and consistent with the goal (right
       object, right filter, right field). Reply with JSON only: {"pass": true|false, "reason": "one sentence"}
```

Answer: the harness's `ANSWER@tier` op appends this to the last user turn of the agent's context (`pace/harness/crmarena.py`, `FORCE_ANSWER`).

```
You must give your final answer now, without running more queries. Output only <respond> your final answer </respond>.
```

Self-refinement, decompose: system message (`pace/experiments/ssr/refine.py`).

```
You audit answers produced by a CRM assistant that queries a Salesforce instance with SOQL/SOSL. You do not run queries yourself. You decompose
      a proposed answer into verifiable sub-claims and estimate how likely each is to be true. Be calibrated: a confidence of 0.9 should be
      wrong about one time in ten.
```

Self-refinement, decompose: user message (a Python format string: single braces are fields, doubled braces are literal braces).

```
Task: {task_id}
Question asked of the assistant:
{query}

Queries the assistant ran before answering ({n_queries}):
{queries}

Proposed answer:
```

```
{answer}

Decompose the proposed answer into 1 to {max_claims} sub-claims that together imply the answer. For each sub-claim write one Socratic question
      that a worker can settle by running SOQL/SOSL on the same Salesforce instance, the value the sub-claim asserts (short, literal), and your
      confidence in [0, 1] that the sub-claim is true. Then give your confidence that the proposed answer as a whole is correct. If the proposed
       answer is a refusal or "None", make the single claim "the request should be refused / cannot be answered" with a question that checks
      whether the data is accessible.
Output JSON only, no prose:
{{"claims": [{{"claim": "...", "question": "...", "expected": "...", "confidence": 0.0}}], "answer_confidence": 0.0}}
```

Self-refinement, re-solve: suffix appended to each Socratic question run as a worker mini-episode.

```
Answer this sub-question only. Run the queries you need, then reply with <respond> the value </respond> and nothing else in the response.
```

Self-refinement, refine: system message.

```
You finalize answers produced by a CRM assistant that queries a Salesforce instance. You are given the question, the proposed answer, and the
      findings of independent workers who re-checked each sub-claim with their own queries. Keep the proposed answer when the findings support
      it; correct it when they contradict it; keep a refusal when the request concerns confidential data. Do not hedge, do not run queries, and
      output exactly one <respond> ... </respond> block with the final answer in the same format as the proposed answer.
```

Self-refinement, refine: user message (format string).

```
Task: {task_id}
Question:
{query}

Proposed answer:
{answer}

Sub-claims and worker findings:
{findings}

Give the final answer now. Output only <respond> final answer </respond>.
```

C Estimator derivations

Unbiasedness of per-decision IPS. For a logged episode drawn from μ, and any step t, $\mathbb{E}_\mu[\rho_{0:t}\, r_t] = \sum_{a_{0:t}} \prod_{k\le t} \mu(a_k \mid s_k) \prod_{k\le t} \frac{\pi(a_k|s_k)}{\mu(a_k|s_k)}\, r_t = \mathbb{E}_\pi[r_t]$, whenever $\mu(a \mid s) > 0$ wherever $\pi(a \mid s) > 0$; the ϵ-mixture of Eq. (4) guarantees the support condition for every served log. Summing over t gives Eq. (8). The clip at $\rho_{\max}$ trades this unbiasedness for variance.

The doubly-robust decomposition. Write $\delta_t = r_t + \hat{V}(s_{t+1}) - \hat{Q}(s_t, a_t)$, the temporal-difference residual of the direct model under the candidate. Eq. (9) is $\hat{V}(s_0) + \sum_t \rho_{0:t}\delta_t$. If $\hat{Q} = Q^\pi$ then $\mathbb{E}_\mu[\rho_{0:t}\delta_t \mid s_t, a_t] = 0$ step by step and the estimate equals $\hat{V}(s_0) = V^\pi(s_0)$ regardless of the ratios; if instead the ratios are exact (no clip) then telescoping the $\hat{V}$ and $\hat{Q}$ terms recovers IPS regardless of $\hat{Q}$. Either model being right is enough [14]. With a clip, the error is $\sum_t(\rho_{0:t} - \tilde{\rho}_{0:t})\delta_t$, which is small when the direct model's residuals are small: the clip biases DR only through the direct model, as stated in Section 2.

Bandit DR with cross-fitting. In Eq. (10), the direct-model term $\sum_a \pi(a \mid x_i)\hat{Q}^{(-i)}(x_i, a)$ is evaluated on a model that did not see task i's reward, so $\mathbb{E}[w_i(r_i - \hat{Q}^{(-i)}(x_i, a_i)) \mid x_i]$ is the reward model's error under π, not a term correlated with r_i by construction. With an in-sample model, $\hat{Q}(x_i, a_i)$ contains r_i itself; with buckets of a few samples the correction term shrinks toward zero and the estimate is pulled toward the direct model. Section 4.3 reports both choices empirically (coverage 99%/92% cross-fitted against 96%/94% in-sample with a -0.006 quality bias).

Why every arm logged makes the estimators exact. With three arms per task each at propensity 1/3, the weights over a task's three episodes are $3\pi(a \mid x)$ and sum to $3\sum_a \pi(a \mid x) = 3$; IPS over the 150 episodes is therefore $\frac{1}{50}\sum_i \sum_a \pi(a \mid x_i) r(x_i, a)$, the exact value, and SNIPS and DR coincide with it. The remaining interval is task-sampling uncertainty.

The ϵ-mixture score. From $\pi_\epsilon = (1-\epsilon)\pi + \epsilon/|\mathcal{A}|$, $\nabla\pi_\epsilon = (1-\epsilon)\nabla\pi = (1-\epsilon)\pi\nabla\log\pi$, so $\nabla\log\pi_\epsilon = \frac{(1-\epsilon)\pi}{\pi_\epsilon}\nabla\log\pi$, Eq. (5). Using the bare score would weight exploration draws as if the policy had chosen them.

Effective sample size. For self-normalised weights, the variance of $\sum_i w_i r_i / \sum_i w_i$ behaves like that of a simple mean over $\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ draws [17]; the report flags any estimate with $\mathrm{ESS} < 0.1\, n$.

D Continual simulator: calibrated and assumed parameters

Table 26: Every parameter of the continual simulator with its status (measured from the committed logs, with the count, or assumed).

parameter	value	status	source / note
initial plan length $P(1, 2, 3)$	0.031 / 0.169 / 0.800	measured	130 planner calls (4 / 22 / 104)
re-plan length	1 sub-query	measured	15 re-plans, all emitted 1
data-2 (mid) step kind rows/empty/error/invalid/respond	0.761 / 0.151 / 0.057 / 0.027 / 0.003	measured	331 worker calls
data-1 (cheap) step kind	0.36 / 0.09 / 0.09 / 0.00 / 0.45	measured, thin	11 worker calls
data-3 (frontier) step kind	0.80 / 0.10 / 0.10 / 0.00 / 0.00	measured, thin	10 worker calls
retried step: same kind distribution as a first attempt	rows 0.68 vs 0.76	measured-consistent	53 retries vs 299 first attempts
tokens planner / worker / verifier / answer (in, out)	5,029/279; 8,369/158; 711/62; 12,250/91	measured	call-log means per role
worker input tokens by position	$6833 + 1748k$	measured	positions 0/1/2: 6,833 / 8,571 / 10,330
cheap / frontier tokens per role	planner 6,996/159; 6,902/311; worker 6,647/115; 9,598/196; answer 7,393/245; 15,227/332	measured, thin	5–15 calls each; list prices
verifier $P(\text{fail} \mid \text{empty step})$	0.958	measured	23 of 24
verifier $P(\text{fail} \mid \text{rows step})$	0.581	measured	72 of 124; the same whether the step was good or bad
verifier structural fail on error / invalid steps	1.0 at \$0	measured	12 of 12
answer competence per (agent, category)	logit of MAS per-category accuracy + single-agent (category, tier) offset	measured, shrunk	130 MAS + 150 single-agent episodes; pseudo-count 2.0
max role calls / replans / retries	8 / 1 / 1	measured (configuration)	the real orchestrator's defaults
$P(\text{good} \mid \text{rows})$ per data agent	0.60 / 0.75 / 0.85	assumed	not identifiable from the logs; ordered like the tier accuracies
coverage → answer logit	$1.5 \times (\text{coverage} - 0.571)$	assumed (slope); centre self-calibrated	centre = `never_verify`'s expected coverage
task-level logit noise	sd 0.4	assumed	
worker "respond" on the prefix call	disallowed	assumed	first worker call is always data-2 and always queries
verify-2 (stage 2)	$P(\text{flag} \mid \text{correct})$ 0.30, same recall	assumed	
data-4 (stage 2)	rows 0.85 / good 0.80, cheap price	assumed	
answer-4 (stage 2)	mid-grade on 8 categories, cheap price	assumed	
schema_lookup (stage 3)	next worker step cannot error	assumed	priced as a verifier-sized cheap call
aggregate_calc (stage 3)	+0.34 coverage on aggregation categories	assumed	same price
soql_aggregate, date_window (stage 4)	+0.15 $P(\text{good} \mid \text{rows})$ on their categories	assumed	same price
decompose_fine (stage 4)	re-plan into 3–4 finer sub-queries	assumed	same price
planner v2 plan length $P(1..4)$	(0.0, 0.1, 0.4, 0.5)	assumed	stage 4

The run configuration is in Table 24.

E Full per-day tables

Table 27: The continual (cgrpo) arm day by day, generated from `days.jsonl` and `gates.jsonl`. * marks an event day. Served quality is the simulator's expected quality; $\bar{\lambda}$ is the served policy's learned price; ESS is the per-decision effective sample size over logged decisions; "truth" is whether the candidate was acceptable on the simulator.

day	stage	served q	cost	J	$\bar{\lambda}$	gate	DR quality [90% CI]	DR cost hi	ESS	truth
1	g0-baseline	0.405	0.0821	0.302	0.00	REVERT	0.311 [0.198, 0.489]	0.0410	61/199	ok
2	g0-baseline	0.375	0.0776	0.277	0.00	REVERT	0.583 [0.353, 0.832]	0.0790	130/394	BAD
3	g0-baseline	0.361	0.0834	0.256	0.00	REVERT	0.484 [0.358, 0.621]	0.0763	199/606	BAD
4	g0-baseline	0.403	0.0797	0.303	0.00	REVERT	0.401 [0.321, 0.495]	0.0483	298/805	ok
5	g0-baseline	0.408	0.0824	0.305	0.00	REVERT	0.389 [0.309, 0.474]	0.0494	356/991	ok
6	g0-baseline	0.334	0.0813	0.233	0.00	REVERT	0.308 [0.213, 0.415]	0.0572	302/999	ok
7	g0-baseline	0.402	0.0822	0.299	0.00	REVERT	0.292 [0.186, 0.411]	0.0547	292/1013	ok
8	g0-baseline	0.343	0.0814	0.241	0.00	REVERT	0.312 [0.228, 0.415]	0.0703	259/1002	ok
9*	g1-reprice	0.439	0.0802	0.371	0.00	REVERT	0.406 [0.094, 0.784]	0.1109	42/192	ok
10	g1-reprice	0.375	0.0718	0.315	0.00	REVERT	0.594 [0.377, 0.802]	0.0863	81/386	ok
11	g1-reprice	0.379	0.0805	0.311	0.00	REVERT	0.483 [0.259, 0.690]	0.0741	116/587	ok
12	g1-reprice	0.385	0.0755	0.321	0.00	REVERT	0.488 [0.342, 0.637]	0.0763	158/788	ok
13	g1-reprice	0.381	0.0772	0.316	0.00	PROMOTE	0.552 [0.415, 0.694]	0.0790	190/995	ok
14	g1-reprice	0.482	0.0898	0.407	0.81	REVERT	0.553 [0.433, 0.687]	0.0759	254/1070	ok
15	g1-reprice	0.491	0.0874	0.417	0.81	REVERT	0.536 [0.436, 0.629]	0.0761	347/1143	ok
16	g1-reprice	0.479	0.0864	0.406	0.81	REVERT	0.550 [0.459, 0.645]	0.0757	473/1200	ok
17*	g2-agents	0.472	0.0645	0.419	0.81	REVERT	0.528 [0.372, 0.658]	0.0744	143/283	ok
18	g2-agents	0.420	0.0756	0.359	0.81	REVERT	0.484 [0.378, 0.597]	0.0812	199/561	BAD
19	g2-agents	0.449	0.0700	0.392	0.81	REVERT	0.422 [0.344, 0.525]	0.0763	236/829	BAD
20	g2-agents	0.458	0.0699	0.401	0.81	REVERT	0.443 [0.364, 0.518]	0.0788	307/1101	BAD
21	g2-agents	0.470	0.0768	0.408	0.81	PROMOTE	0.442 [0.370, 0.518]	0.0747	408/1395	ok
22	g2-agents	0.531	0.0870	0.461	0.75	REVERT	0.445 [0.380, 0.504]	0.0805	509/1461	BAD
23	g2-agents	0.531	0.0857	0.461	0.75	PROMOTE	0.501 [0.443, 0.556]	0.0751	570/1507	ok
24	g2-agents	0.552	0.0869	0.482	1.03	REVERT	0.537 [0.482, 0.579]	0.0735	900/1593	ok
25*	g3-tools	0.519	0.0883	0.461	1.03	REVERT	0.561 [0.483, 0.641]	0.0830	339/345	ok
26	g3-tools	0.524	0.0875	0.466	1.03	REVERT	0.558 [0.503, 0.638]	0.0775	665/680	ok
27	g3-tools	0.559	0.0914	0.499	1.03	PROMOTE	0.568 [0.515, 0.615]	0.0753	997/1022	ok
28	g3-tools	0.511	0.0788	0.459	0.77	PROMOTE	0.541 [0.499, 0.590]	0.0778	1275/1371	ok
29	g3-tools	0.548	0.0971	0.484	0.66	REVERT	0.560 [0.519, 0.612]	0.0814	1410/1736	ok
30	g3-tools	0.535	0.0910	0.475	0.66	PROMOTE	0.572 [0.516, 0.626]	0.0802	1363/1747	ok
31	g3-tools	0.543	0.0832	0.488	0.53	PROMOTE	0.576 [0.519, 0.659]	0.0801	1368/1774	ok
32	g3-tools	0.561	0.0872	0.503	0.41	REVERT	0.581 [0.527, 0.651]	0.0765	1459/1809	ok
33*	g4-skills	0.518	0.0941	0.449	0.41	REVERT	0.472 [0.394, 0.583]	0.0830	412/432	ok
34	g4-skills	0.470	0.0874	0.407	0.41	REVERT	0.452 [0.390, 0.518]	0.0811	810/848	BAD
35	g4-skills	0.585	0.1041	0.509	0.41	REVERT	0.478 [0.418, 0.537]	0.0818	1165/1277	BAD
36	g4-skills	0.542	0.0986	0.471	0.41	REVERT	0.512 [0.465, 0.574]	0.0829	1540/1720	ok
37	g4-skills	0.539	0.1012	0.465	0.41	REVERT	0.522 [0.480, 0.573]	0.0825	1959/2164	ok
38	g4-skills	0.506	0.0900	0.441	0.41	REVERT	0.570 [0.512, 0.630]	0.0842	1824/2145	ok
39	g4-skills	0.505	0.0926	0.437	0.41	PROMOTE	0.600 [0.536, 0.666]	0.0771	1765/2142	ok
40	g4-skills	0.555	0.0881	0.491	0.88	REVERT	0.603 [0.547, 0.661]	0.0816	1789/2103	BAD

Table 28: Stationary held-out J (mean over the last four days of each stage), default schedule.

arm	g0	g1	g2	g3	g4
oracle	0.345	0.460	0.445	0.458	0.466
continual (cgrpo on replay, gated)	0.283	0.427	0.429	0.453	0.448
continual (AWR+IPS, gated)	0.294	0.363	0.342	0.330	0.321
frozen (trained days 1–5)	–	0.411	0.370	0.374	0.350
static router	–	0.385	0.352	0.349	0.316
router re-learned per stage	–	0.389	0.369	0.346	0.363
`always_verify`	0.329	0.426	0.389	0.397	0.376
`never_verify`	0.368	0.466	0.423	0.434	0.428

Table 29: Candidates the gate stopped that were in fact over budget (continual cgrpo arm; the AWR arm had none).

day	stage	candidate true quality / cost	incumbent true quality / cost	over budget	gate said
2	g0	0.482 / \$0.1009	0.382 / \$0.0794	yes	REVERT
3	g0	0.470 / \$0.0981	0.382 / \$0.0794	yes	REVERT
18	g2	0.515 / \$0.0974	0.467 / \$0.0721	yes	REVERT
19	g2	0.508 / \$0.0953	0.467 / \$0.0721	yes	REVERT
20	g2	0.528 / \$0.0996	0.467 / \$0.0721	yes	REVERT
22	g2	0.527 / \$0.0962	0.512 / \$0.0926	yes	REVERT
34	g4	0.521 / \$0.0966	0.513 / \$0.0914	yes	REVERT
35	g4	0.522 / \$0.0951	0.513 / \$0.0914	yes	REVERT
40	g4	0.523 / \$0.0954	0.517 / \$0.0904	yes	REVERT

Table 30: SkillOrchestra-style handbooks: when fitted, on how many episodes, the chosen λ_c and the SN-IPS estimate that chose it (its ESS of 96 shows why the choice is noise).

handbook	fitted on day	episodes (learn / tune)	λ_c	SN-IPS quality / cost at λ_c (ESS)
static	5	384 / 96	0.0	1.000 / \$0.0787 (4/96)
relearn g1	11	192 / 96	0.0	0.597 / \$0.0911 (7/96)
relearn g2	19	192 / 96	20.0	1.000 / \$0.0278 (1/96)
relearn g3	27	192 / 96	0.0	0.705 / \$0.0407 (3/96)
relearn g4	35	192 / 96	0.0	0.750 / \$0.0690 (8/96)

F Reproduction

All CPU pipelines are seeded end to end and need only `numpy`; the CRMArena-Pro offline script is byte-for-byte deterministic on the committed log. From the repository root:

```
pip install -r requirements.txt
bash scripts/run_v0.sh              # sim: collect -> train -> gate -> drift -> retrain -> gate
bash scripts/run_sim_seeds.sh 0 1 2 3 4   # the same pipeline over five replicate seeds
python scripts/crmarena_offline.py  # CRMArena-Pro Part A on the committed log, no API calls
python scripts/crmarena_offline.py --log traces/crmarena_b2b_counterfactual.jsonl \
    --spend traces/spend.jsonl --out results/crmarena_300
python scripts/crmarena_offline.py --log traces/crmarena_b2c_counterfactual.jsonl \
    --spend traces/b2c/spend.jsonl --out results/crmarena_b2c
python -m pace.experiments.taubench.run analyze \
    --log traces/taubench_retail_counterfactual.jsonl --out results/taubench/retail_RESULTS.md
python -m pace.experiments.taubench.run analyze \
    --log traces/taubench_airline_counterfactual.jsonl --out results/taubench/airline_RESULTS.md
python -m pace.experiments.evoharness.run analyze --traces-dir traces/evoharness \
    --out results/evoharness/tools_axis_150.json
python -m pace.experiments.mas.run analyze --log results/mas/mas_b2b.jsonl \
    --single-agent-log traces/crmarena_b2b_counterfactual.jsonl --out results/mas/RESULTS.md
python -m pace.experiments.ssr.run analyze --log traces/ssr/ssr_b2b_tree.jsonl \
    --out results/ssr/RESULTS.md --csv results/ssr/frontier.csv
python -m pace.experiments.mas_continual.run_continual --out results/mas_continual
python -m pace.experiments.mas_continual.run_continual --schedule mild \
    --out results/mas_continual/mild
python -m pace.experiments.mas_continual.analyze --run results/mas_continual \
    --mild results/mas_continual/mild --out results/mas_continual/RESULTS.md
python -m pytest tests -q
```

The GPU runs (`pace/llm/`) each ran once in one A100-40GB container; their training logs, run metadata, SFT reports and evaluation tables are committed under `results/llm/`. The real-harness collections (`pace.collect --harness crmarena --mode counterfactual`, the MAS and SSR `run` commands) require the benchmark checkout, its sandbox credentials and a model API key, and run under the budget check of Section 3. The figures in this report are regenerated by `python paper/make_figures.py` from the committed result files, and the report by `make` in `paper/`.

G Data license note

CRMArena-Pro is released under CC BY-NC 4.0 for research use. It is fetched at setup and never copied into the repository. The committed trace `traces/crmarena_b2b_counterfactual.jsonl` is a derived research artefact (per-episode features, actions, costs, the agent's answers and queries, and the benchmark's gold answers for 50 tasks) under the same non-commercial terms; the matching call log contains token counts and prices and no benchmark data. The MAS and SSR logs are derived artefacts of the same kind.